



# Absolute Calibration. III. Improved Absolute Calibration for the Visible through the Mid-infrared

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## Abstract

We propose Sirius as an improved zero-point-defining star and calibrate its spectrum to an accuracy of  $\sim 0.6\%$  in both the visible and infrared. This result is based on a newly derived independent calibration in the visible of similar accuracy to the previous standard one, with which it is combined. We use a large variety of approaches in the infrared to reach about three times smaller error than for previous absolute calibrations. The results in the two wavelength regimes are in agreement, providing a consistent link from the visible throughout the near- and mid-infrared. The Sirius-based zero-point at 5557.5 Å (in vacuum) is  $13.436 \pm 0.081 \times 10^{-12} \text{ W cm}^{-2} \mu\text{m}^{-1}$ , based on the improved value for Vega of  $3.473 \pm 0.018 \times 10^{-12} \text{ W cm}^{-2} \mu\text{m}^{-1}$  and the measured magnitude difference between the two stars. At  $2.1603 \mu\text{m}$ , the zero-point is  $4.225 \pm 0.025 \times 10^{-14} \text{ W cm}^{-2} \mu\text{m}^{-1}$  taking Sirius at a magnitude of  $-1.395$ . A jackknife analysis indicates that there are no serious systematic errors in these results. We consider selection of secondary standards that can extend the calibration over the sky. Despite more than a century in this role, normal A-stars are not suitable, although Am and Ap stars may be. G-stars older than  $\sim 1$  Gyr are good candidates if accurate temperatures can be measured. White dwarfs are suitable from the visible through the near-infrared, but their properties are unexplored at the necessary level at the longer infrared wavelengths, and for most facilities they are too faint there. Finally, as a further test of the calibration, we demonstrate an upgraded infrared flux method to determine accurate stellar diameters from *K*-band photometry.

*Unified Astronomy Thesaurus concepts:* Infrared photometry (792); Calibration (2179)

*Supporting material:* machine-readable tables

## 1. Introduction

Many astronomical observations can be interpreted on a relative basis through comparison of different sources measured in identical ways. This approach has been the foundation, for example, of stellar photometry. However, interpreting astronomical observations physically generally requires that they be expressed in physical units. An accurate conversion of astronomical observations to physical units—an absolute calibration—is therefore critical for many aspects of astronomy. Some examples are:

1. Such calibrations can be essential to combine measurements obtained with different instruments in a consistent way, e.g., in constructing a spectral energy distribution (SED) bridging from the visible<sup>6</sup> to the infrared and, indeed, across the infrared where different instruments are employed for different wavelength regimes. Calibration uncertainties represent systematic errors that cannot be reduced by obtaining measurements of intrinsically high signal to noise. Thus, for studies requiring high accuracy over multiple wavelength regimes, the absolute calibrations in those regimes need to be both consistent

with each other and more accurate than the random measurement uncertainties to achieve high fidelity SEDs.

2. As an important example, measuring the expansion of the universe using Type Ia supernovae requires an accurate cross-calibration between the visible and the infrared to provide consistent results as a given wavelength regime passes from the visible to the infrared with increasing redshift. This issue is discussed in depth by Kent et al. (2009), who conclude that a calibration consistent to  $\lesssim 1\%$  is needed across this boundary.
3. Infrared calibrations have a special role in stellar studies, since the behavior of stars is relatively simple in the Rayleigh–Jeans portion of their spectra. The infrared flux method (IRFM) takes advantage of this attribute to use infrared photometry and bolometric flux measurements of stars to estimate accurate angular diameters. This procedure is a sensitive test of stellar emission models, helping to constrain fundamental parameters such as effective temperatures, surface gravities, and composition. The IRFM can also be used where stellar diameters are themselves of interest, for example for estimating exoplanet diameters from transit observations. The systematic error in the derived diameters goes as the square root of the uncertainty in the absolute calibration. Given other contributions to the error budget (e.g., uncertainties in the stellar temperatures), an infrared calibration accurate to  $\lesssim 1\%$  is needed to estimate stellar angular diameters to  $\sim 1\%$ .

A calibration in the visible band with an accuracy of  $\sim 0.7\%$  was reported some time ago (Megessier 1995). The CCDs in

<sup>6</sup> We use the common terminology that the visible includes wavelengths from 0.4–0.8  $\mu\text{m}$ , with longer wavelengths falling in the infrared.



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STIS on the HST have the property that when the signal from a bright source overflows the CCD wells, the signal appears in surrounding pixels without significant loss (Bohlin & Gilliland 2004). This attribute has allowed Ralph Bohlin and collaborators to extend this calibration to a network of faint,<sup>7</sup> accurately calibrated stars in the visible and ultraviolet. Even in this situation, however, confirming the desired 1% accuracy is very challenging and, for example, efforts continue to improve both the understanding of the CCD performance and also models of the calibrator stars (e.g., Bohlin et al. 2020).

Infrared calibrations have not yet approached the  $\lesssim 1\%$  uncertainty range. There have been relatively few high-accuracy direct absolute measurements (i.e., traceable directly to calibrated sources) and therefore intercomparison of independent measurements has been limited, making it difficult to test for systematic errors. Current calibrations rely strongly on the results from the Mid-Course Space Experiment (MSX; Price et al. 2004a). A second avenue for accurate calibration relies on the absolute solar infrared flux, but until recently there have been significant discrepancies among the available measurements of the solar flux in the infrared (e.g., Gueymard 2018). There has also been no direct way to translate the achieved direct calibrations to faint stars, other than through a chain of photometric measurements that may impress additional uncertainties on the results. In part because of concerns about systematic errors, the available infrared calibrations have quoted uncertainties of  $\sim 2\%$  (e.g., Rieke et al. 2008; Engelke et al. 2010; Carey et al. 2012; Sloan et al. 2015).

A number of advances now make possible significant improvements in the absolute calibration in both the visible and infrared. Hipparcos has provided an all-sky visible wavelength set of stellar photometry accurate to better than 1%. The issues with measuring the infrared flux from the Sun have been largely resolved (Gueymard 2018; Coddington et al. 2021). Accurate diameters have been determined for many stars with interferometers (e.g., Boyajian et al. 2013), opening up new possibilities for quantitative comparisons of their outputs. Su et al. (2022) developed a method for direct transfer in the infrared from very bright stars to ones within reach of normal photometric networks, allowing a check of the photometric chains used for this purpose previously.

This paper takes advantage of these advances to improve the accuracy of the visible absolute calibration to 0.5%–0.6% and the infrared absolute calibration from  $\sim 2\%$  to  $\sim 0.6\%$ . The work emphasizes multiple approaches to guard against systematic effects in any one of them: (1) for the visible, we unite measurements of calibrated laboratory sources by four groups and of multiple stars, plus we carry out a fifth approach based on absolutely calibrated measurements of the Sun; and (2) the infrared calibration is based on eight different approaches. Our results therefore represent a significant advance.

We show that the spectrum of Sirius provides a consistent link between the visible and infrared and can therefore provide an improved definition of the zero-point for stellar photometry. It is linked accurately to existing visible standard star networks down to faint white-dwarf stars (Bohlin et al. 2020). In the infrared, the detailed photometric analysis in Rieke et al. (2022) is a significant step toward transferring the calibration from Sirius to stars sufficiently faint for standard photometry. One of our calibration approaches in the infrared uses stars with

**Table 1**  
Calibration of Vega at 5557.5 Å

| reference                           | $\text{W cm}^{-2} \mu\text{m}^{-1}$ | error ( $\text{W cm}^{-2} \mu\text{m}^{-1}$ ) |
|-------------------------------------|-------------------------------------|-----------------------------------------------|
| Hayes & Latham (1975)               | $3.45 \times 10^{-12}$              | $0.069 \times 10^{-12}$                       |
| Tüg et al. (1977)                   | $3.47 \times 10^{-12}$              | $0.035 \times 10^{-12}$                       |
| Terez (1977)                        | $3.44 \times 10^{-12}$              | $0.041 \times 10^{-12}$                       |
| Megessier average                   | $3.457 \times 10^{-12}$             | $0.025 \times 10^{-12}$                       |
| Additional Measurements             |                                     |                                               |
| Tüg et al. (1977) via 109 Vir       | $3.50 \times 10^{-12}$              | $0.052 \times 10^{-12}$                       |
| Tüg (1980a)                         | $3.50 \times 10^{-12}$              | $0.042 \times 10^{-12}$                       |
| Terez & Terez (1979)                | $3.50 \times 10^{-12}$              | $0.070 \times 10^{-12}$                       |
| Willstrop (1960)                    | $3.46 \times 10^{-12}$              | $0.070 \times 10^{-12}$                       |
| from V of Sun                       | $3.465 \times 10^{-12}$             | $0.070 \times 10^{-12}$                       |
| average for additional measurements | $3.490 \times 10^{-12}$             | $0.025 \times 10^{-12}$                       |
| grand average                       | $3.473 \times 10^{-12}$             | $0.018 \times 10^{-12}$                       |
| equivalent for Sirius <sup>a</sup>  | $13.436 \times 10^{-12}$            | $0.081 \times 10^{-12}$                       |

**Note.**

<sup>a</sup> This is the flux density of Sirius assuming the measured magnitude difference between it and Vega.

accurate diameters measured with interferometry to determine the expected fluxes in the infrared. The success of this approach provides a basis for improving the use of infrared photometry for estimating stellar diameters. That is, the improved calibration provides a significant advance toward the goals itemized above.

Those wanting to scan our results can consult Tables 1 and 3 and read Sections 4 and 5. The details of the visible calibration are described in Section 2 and of the infrared one in Section 3. Section 4 covers some implications of this work, including tests for systematic errors in the calibration, an analysis of strengths and issues in types of secondary calibrator stars, and an evaluation of use of infrared photometry in determining stellar diameters. A succinct summary is provided in Section 5. This paper is also a key step toward the accurate absolute calibration of the JWST (Gordon et al. 2022).

## 2. Calibration in the Visible

For nearly 30 yr, the absolute calibration in the visible, referred to a wavelength (vacuum) of 5557.5 Å, has been taken from the review by Megessier (1995). This situation reflects both the quality of the review and the absence of relevant recent measurements. She combined three independent measurements relative to calibrated sources, respectively, by Hayes & Latham (1975), Tüg et al. (1977), and Terez (1977). The values are listed in Table 1. However, the availability of all-sky photometry of high stability and accuracy from Hipparcos opens up possibilities to improve upon this result, as discussed in this section. The augmented calibration is shifted  $\sim 0.3\%$  bright relative to that from Megessier (1995), with a nominal error of 0.49%, reduced from 0.72%. All of the inputs to this final value are listed in Table 1, and are discussed below. The table also shows the equivalent flux density for the proposed Sirius-based zero-point.

<sup>7</sup> V magnitudes  $\sim 15$ .

### 2.1. Calibrations Using Standard Sources

The three calibrations used by Megessier (1995) were all based on direct transfers from standard flux sources to Vega. The star 109 Vir has been promoted as an alternative calibration-defining star. Measurements of it by Tüg et al. (1977) can provide a fourth calibration. Tüg (1980a, 1980b) obtained absolute flux measurements of a number of southern stars using the same procedures applied to Vega and 109 Vir. Some of these stars are poorly chosen (e.g., they are variable), but seven of them are suitable for transfer of the calibration to Vega: HR 591, 721, 1347, 6165, 7790, 8425, and 8937. Another set of measurements is described by Terez & Terez (1979). We will make use of the two stars (other than Vega) for which they obtained two independent sets of measurements at different observatories, and hence which should have the smallest internal errors:  $\alpha$  Oph and  $\alpha$  Peg.

Since all of these measurements are by the same groups using the same equipment as the direct calibrations of Vega, transferring them to the calibration of Vega should reduce random errors and also help identify systematic ones to the extent that they are different among the groups. To transfer from the other stars to Vega, we utilize the Hipparcos  $H_p$  magnitudes, which are accurate to significantly better than 1% and are homogeneous over the sky. We transform them to  $V$  as derived by Harmanec (1998). Sadly, Vega is just bright enough that its  $H_p$  magnitude is undermined by saturation (Bohlin 2014), so to make use of these  $V$  magnitudes, it is necessary to put them on a common scale with Vega itself. We do this in two ways.

Our first transfer to this scale uses the CALSPEC database (Bohlin et al. 2020). It contains spectra of a number of A-type stars, including Vega, all on the same scale due to the property of the STIS CCDs that they preserve electrons even when the CCD wells overflow. To compare the  $V$ -band brightnesses of A-type stars in this database, we carry out synthetic  $V$  photometry<sup>8</sup> on the spectra of Vega and the other stars to determine the  $V$  magnitude differences. We then compare the results with  $V$  magnitudes of the same stars derived from the  $H_p$  ones. To provide the highest possible accuracy, we focus just on stars of types A0V and A1V,<sup>9</sup> to minimize  $B - V$  and  $U - B$  and hence the color corrections in the transformation from  $H_p$  to  $V$ . Comparing with the  $H_p$ -based  $V$  values indicates that the Hipparcos-based values need to be adjusted brighter by 0.021 mag to match them to Vega at its standard  $V$  magnitude of 0.033 (Mermilliod 1991).

A second determination is based on the photometrically measured  $V$  magnitudes. We start with the bright star photometry of Johnson et al. (1966). We select from this compilation stars with  $-0.2 < B - V < 0.2$  (to minimize color terms in the photometric transformation), with many repeat measurements (designated as “9” in the key), main sequence, and with  $V$  magnitude  $> 2$  (to stay well away from saturation with Hipparcos), and with no indication of variability in the literature.<sup>10</sup> Since these stars are measured in a stable photometric system, the relative  $V$  magnitudes yield an accurate comparison with Vega (which is not saturated in the Johnson

photometry). These comparison stars are also measured at high accuracy with Hipparcos/Tycho. We compare the two  $V$  magnitudes for each star, finding that the Hipparcos-based ones need to be made brighter by 0.011 mag to be on the Vega scale. We repeated this sequence using the compilation of magnitudes by Mermilliod (1991) and found the identical result.

From these two determinations, we make an adjustment of 0.016 mag brighter in the Hipparcos-based  $V$  magnitudes to relate them to Vega, with an uncertainty of 0.005 mag. We apply this adjustment to relate the absolutely calibrated measurements of all of the stars mentioned in the first paragraph of this section to derive independent estimates of the absolute flux from Vega. The results can be found in Table 1. We list 109 Vir separately because it was measured in the same series with Vega, then those from the southern campaign of Tüg (1980a, 1980b), and then the result from Terez & Terez (1979); we assign errors of 1.5% to the result for 109 Vir, 1.2% to the average the seven stars from Tüg (1980a, 1980b), and 2% for Terez & Terez (1979).<sup>11</sup> All six determinations of the flux of Vega from calibrated sources are consistent with each other within the errors, indicating no major systematic errors imposed by the different reference source types and calibrations.

There is also an absolute measurement of Sirius itself by Willstrop (1960). We have adjusted his  $y$  photometric system to true  $V$  magnitudes by comparing stars with  $y - b_1 \sim 0.14$  (the value for Sirius) with  $V$  magnitudes from Mermilliod (1991). As a result, we make an adjustment of 0.016 mag brighter in his measurement of Sirius. We then transfer from the measured wavelength of 5390 Å to 5557.7 Å according to the CALSPEC Sirius spectrum; his measurement then becomes  $1.336 \pm 0.026 \times 10^{-12} \text{ W cm}^{-2}$ . Although our final goal is a flux of Sirius, to compare with the ones for Vega we proceed as follows. As shown in Figure 1, we have fitted the stellar continua for both stars with power laws from 5450–5700 Å and with quartics from 5400–6200 Å (both covering ranges of clean continuum)<sup>12</sup> and at 5557.5 Å both indicate a difference between the stars of 1.469 mag.<sup>13</sup> Applying this difference to the measurement by Willstrop (1960), we obtain  $3.46 \times 10^{-12} \text{ W cm}^{-2} \mu\text{m}^{-1}$ ; he estimates the error to be 2%.

### 2.2. Calibration Via the Sun

An additional calibration is possible based on observations that relate the stellar  $V$  magnitude system directly to the  $V$  magnitude of the Sun. The measurements are listed in Table 2, along with an additional one of historic interest (which is not used in this analysis). Lockwood et al. (1992) used a diffractive device to make a direct comparison between Vega and the Sun.

<sup>8</sup> Here, and throughout this paper, synthetic photometry is conducted on the relative spectral response function across the relevant photometric band, matched carefully to the observed band including the effects of all active spectral elements, i.e., on the system photon conversion efficiency in that band as a function of wavelength.

<sup>9</sup>  $\alpha$  Lyr,  $\delta$  UMi,  $\epsilon$ 1 Dor, HD 116405, HD 128998, and HD 180609.

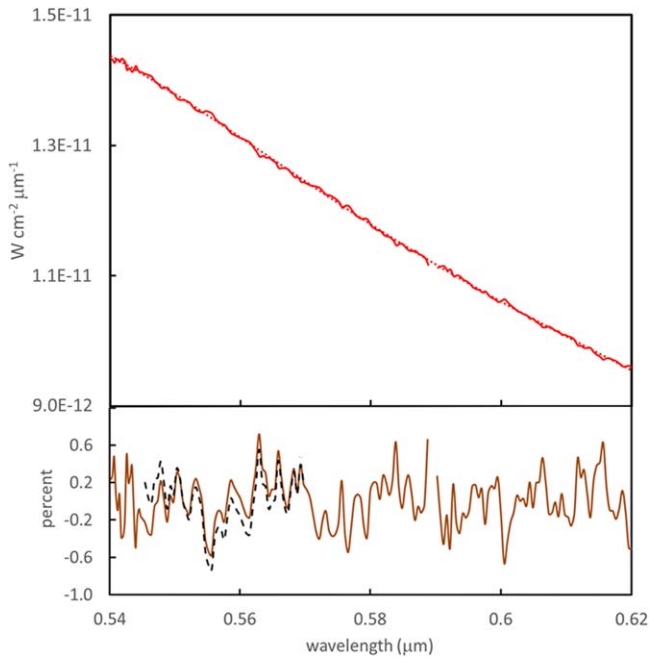
<sup>10</sup> 109 Vir, HR 343, 553, 875, 1172, 1641, 1938, 1998, 2763, 3226, 3314, 3569, 3665, 4456, 5359, 5531, 5685, 5867, 6092, 6175, 6629, 7001, 7235, 7767, 7906, 7950, 8622, 8709, and 8781.

<sup>11</sup> For 109 Virgo, Tüg et al. (1977) quote an error of 1%, and we add 0.5% for the systematic error in tracing the Hipparcos magnitude to the  $V$  magnitude of Vega. Tüg (1980a) reports errors similar to those for Vega, i.e., 1% in the spectral region of interest. There are seven stars. Some of the listed error terms will be reduced in the average for these stars; calculating the net error allowing for this effect makes it 0.7%–0.8%. Again we have 0.5% for the tracing to  $V$ . Terez & Terez (1979) quote a typical error for the stars they measure of  $\sim 2\%$  at the wavelengths of interest. Since we have taken the two best measured stars, each observed independently at two observatories, we take the error on the average to be 1.5% and again add 0.5% for the systematic error.

<sup>12</sup> See Section 2.3 for discussion of this approach.

<sup>13</sup> The  $V$  magnitude of Sirius is determined at  $-1.440 \pm 0.009$  by Turon et al. (1992), i.e., a difference relative to Vega of 1.473 mag, which represents the best value from historic photometry. Averaging over 25 Å windows centered on 5557.5 Å, Bohlin gets a difference of 1.471 mag. Both values agree satisfactorily with our proposed difference, although our determination should be more robust.





**Figure 1.** Top: quartic fit to the spectrum of Sirius; the solid line is the spectrum and the dotted one the fit. The short gap in the spectrum is where a spectral feature has been masked. Bottom: residuals in the fit, shown as fit—spectrum. The quartic fit is shown as a solid line and a power-law fit as a dashed one.

Although they described this measurement as a means to determine the absolute flux from the Sun, we have inverted it and used the very well-measured flux from the Sun from Coddington et al. (2021) to obtain a measurement of the flux of Vega. Lockwood et al. (1992) estimated the 95% confidence errors in their measurement to be 2%; the error in the solar flux is estimated at no more than 0.25%. We adopt 2% as the  $1\sigma$  error. Another calibration along similar lines uses the comparison of the Sun with stars by Stebbins & Kron (1957) as discussed by Hayes (1985), who improved the estimate of atmospheric extinction and derive a revised value with an estimated error of 3%. A third measurement is reported by Gallouët (1964), who related 12 Johnson-Morgan primary calibration stars to the Sun, using a system of convex mirrors and diaphragms to reduce the flux from the Sun. He claimed an accuracy of 1% overall, which we increase to 2% to be conservative, and so this one measurement does not completely dominate the weighted average of the three.

Comparing these measurements with the calibration of the A-type stars cannot be done by taking the flux at a fiducial wavelength, both because of the overall differences between the A-star and G-star spectra, and because the complex of absorption lines in a G-star spectrum result in it not having a well-determined smooth continuum to the necessary level of accuracy. Instead, we have compared the synthetic  $V$  magnitudes of the Sun and Vega, the first based on the spectrum of Coddington et al. (2021) and the second from CALSPEC. The entry in Table 1 is based on this comparison, with the flux of Vega at 5557.5 Å adjusted to match the observed difference in  $V$ -band photometry. As indicated, the agreement is nearly perfect. Viewed in this fashion there is a 2% uncertainty in the calibrated flux of Vega from this one approach. However, given the overall accuracy of the calibrations of both Vega and of the Sun, one can turn the

**Table 2**  
 $V$  Magnitude of the Sun

| Reference                     | Magnitude           | Error (Magnitude) |
|-------------------------------|---------------------|-------------------|
| Nikinov (1949) <sup>a</sup>   | −26.81              | 0.07              |
| Kariagina (1955) <sup>a</sup> | −26.83              | 0.06              |
| Stebbins & Kron (1957)        | −26.75 <sup>b</sup> | 0.03              |
| Gallouët (1964)               | −26.70              | 0.01 <sup>c</sup> |
| Lockwood et al. (1992)        | −26.75              | 0.02              |
| Willmer (2018) <sup>d</sup>   | −26.76              | 0.03              |
| average                       | −26.740             | 0.02 <sup>e</sup> |
| Historical Interest           |                     |                   |
| Pettit & Nicholson (1928)     | −26.88              | 0.09              |

**Notes.**

<sup>a</sup> From Johnson (1965).

<sup>b</sup> As corrected by Hayes (1985).

<sup>c</sup> Increased to 0.02 for weighted average.

<sup>d</sup> This value is based on a critical analysis of other measurements rather than being a direct measurement.

<sup>e</sup> Nominal weighted average error is 0.012. However, this does not reflect possible systematic errors. To test for them, we eliminated Gallouët (1964; since it is the lowest value and has the highest weight) and Willmer (2018; which is not strictly an independent measurement), which changes the average to −26.76; we then decreased the error for Gallouët (1964) to 0.01, which changes it to −26.72. Eliminating the two highest values has little effect, given their low weights. Therefore, an error of 0.02 has been adopted.

argument around to show that the  $V$  magnitude of the Sun is  $-26.74 \pm 0.01$ .

### 2.3. Comment

Using the calibration to the quoted accuracy requires a standard procedure to interpret stellar measurements. Specifically, there is a risk that a spectral feature might fall right on the fiducial wavelength for a calibration. The calibration measurements themselves are generally made at modest spectral resolution, i.e., 45 and 30 Å (Hayes 1970),  $\sim 50$  Å (Oke & Schild 1970), 50 Å (Terez 1977), and 10 and 20 Å (Tüg et al. 1977; Tüg 1980a). Hayes (1985) suggested standardizing on quoting fluxes for the average over a 25 Å range centered on the quoted fiducial wavelength. However, at the level of accuracy now being achieved, this approach can produce some apparently arbitrary small offsets due to noise and other artifacts. For example, in comparing the CALSPEC STIS spectrum of Sirius and the model of Sirius, both normalized to perfect agreement in the infrared, a 25 Å window gives a difference of  $\sim 0.6\%$  and a 100 Å window gives one of  $\sim 0.35\%$ . Better agreement can be achieved by fitting a power law to the clean, continuum-dominated spectrum between 5450 and 5700 Å, or a quartic between 5400 and 6350 Å, both of which suggest a difference of 0.2% (with the STIS spectrum, the brighter). The continuum fits provide an optimal smoothing over artifacts and make use of a larger spectral range than the 25 Å window, so we prefer them as the most accurate way to estimate the true flux. However, as discussed in Section 2.2, this approach breaks down with later-type stars with complex absorptions in their spectra, and for them the calibration is most readily conveyed from one star to another using synthetic photometry on the spectra. Indeed, even for the A- and G-type stars, there can be offsets at the 0.1%–0.2% level, so the

**Table 3**  
Summary of Calibrations: Sirius at 2.1603  $\mu\text{m}$

| Method <sup>a</sup>       | $\text{W cm}^{-2} \mu\text{m}^{-1}$ | Error ( $\text{W cm}^{-2} \mu\text{m}^{-1}$ ) | Wavelengths ( $\mu\text{m}$ ) |
|---------------------------|-------------------------------------|-----------------------------------------------|-------------------------------|
| 1. MSX (3.2.2)            | $15.20 \times 10^{-14}$             | $0.21 \times 10^{-14}$                        | 8, 12, 15, 21                 |
| 2. Solar (3.3)            | $15.23 \times 10^{-14}$             | $0.18 \times 10^{-14}$                        | 2.16                          |
| 3. Near Infrared (3.4)    | $15.63 \times 10^{-14}$             | $0.31 \times 10^{-14}$                        | 2.2, 2.25, 3.76               |
| 4. Viking IRTM (3.5)      | $15.53 \times 10^{-14}$             | $0.51 \times 10^{-14}$                        | 11                            |
| 5. Inverted IRFM (3.6)    | $15.00 \times 10^{-14}$             | $0.23 \times 10^{-14}$                        | 3.55                          |
| 6. Solar Analog (3.7)     | $15.32 \times 10^{-14}$             | $0.37 \times 10^{-14}$                        | 0.56, 2.16                    |
| 7. Sirius SED (3.8)       | $15.33 \times 10^{-14}$             | $0.23 \times 10^{-14}$                        | 0.55575                       |
| 8. White Dwarf (3.9)      | $15.29 \times 10^{-14}$             | $0.31 \times 10^{-14}$                        | 0.55575                       |
| average                   | $15.26 \times 10^{-14}$             | $0.09 \times 10^{-14}$                        |                               |
| zero-point <sup>b</sup>   | $4.222 \times 10^{-14}$             | $0.025 \times 10^{-14}$                       |                               |
| diameter of Sirius (3.10) | $15.40 \times 10^{-14}$             |                                               |                               |

**Notes.**

<sup>a</sup> Each method is followed by the paper section where it is discussed.

<sup>b</sup> Taking the magnitude of Sirius to be  $-1.395$

ultimate accuracy will always be to take the calibrated spectrum of Sirius and use it to synthesize the system response.

### 3. Infrared-based Calibrations

We now turn to calibrations from 2–25  $\mu\text{m}$ . We emphasize a variety of approaches to identify any significant systematic errors. To allow for comparisons, we have transferred each of the calibrations from its original wavelength to the pivot wavelength (Bohlin et al. 2014) of the Two Micron All Sky Survey (2MASS)  $K_S$  band, 2.1603  $\mu\text{m}$ . To do so, we have normalized the 2020 CALSPEC spectrum of Sirius (Bohlin et al. 2020) to a given measurement and then determined the flux of the normalized spectrum of Sirius at the  $K_S$  pivot wavelength. We interpolate the Sirius spectrum over the Br  $\gamma$  absorption feature for this determination, using a power law fitted between 2.00 and 2.31  $\mu\text{m}$  (that is over the FWHM of the  $K_S$  passband), after masking the Br  $\gamma$  absorption line.

We will focus throughout this discussion on this calibration of the Sirius SED, i.e., on the proposed zero-point for stellar photometry. For reference, the magnitude of the star on the 2MASS system and at this wavelength is  $-1.395$  (Rieke et al. 2022). We first describe five calibrations traceable directly to a calibrated infrared source, and then three more where the connection to such a source is more indirect. Our progress through these different approaches can be followed in Table 3.

#### 3.1. Calibration Conventions

The various absolute calibrations we discuss below are based primarily on photometry with bandpass filters of significant width in wavelength. It has been conventional in this situation to convert a calibration made for a given filter to a flux density at a specific wavelength based on one of a number of possible definitions of the optimum wavelength to use with the filter (e.g., mean, nominal, isophotal, or pivot wavelength). At the accuracy goals for this paper, this approach is no longer sufficient. As an example, for the discussion in Section 3.9, we compare synthetic photometry and continuum fits of white-dwarf stars and Sirius in the two short wavelength IRAC bands. There are differences up to the 0.5% level in the comparison. These differences arise because of the absorption features in the spectrum of Sirius, compared with the nearly complete absence

of such features for the white dwarfs; otherwise, the continua are very similar, i.e., nearly Rayleigh–Jeans. Basing the calibration on the synthetic photometry therefore makes it specific to the spectral band and not directly related to other photometric systems. To avoid this issue, we will compare stellar continua, determined by fitting power laws to the spectra with the absorption features masked (if necessary, in an iterative process). These fits converge accurately and with only two free parameters (normalization and slope). Given that the infrared fluxes are all in the Rayleigh–Jeans regime for the calibration stars, with only a modest number of absorption features in the spectra, this characterization gives a unique and well-determined result.

The comparison of continua provides a useful general approach. However, to retain the full inherent accuracy of the absolute calibration in more specific applications, convolving the SED of the calibration star and that of the target source with the observing system relative response function and integrating the predictions of the system responses to the two sources is the optimum procedure.

#### 3.2. The MSX Calibration

##### 3.2.1. MSX Bright Star Photometry

The classical absolute calibration in the infrared is based on measurements by the SPIRIT III instrument on MSX, as discussed by Price et al. (2004a). There are two efforts reported in that paper. The first is a set of photometry in various infrared bands of a set of bright stars. The results are quoted in terms of the ratios to calibrated stellar templates derived by Cohen et al. (1995, 1996). The second is an observation of calibrated emission spheres ejected by the spacecraft, used to achieve an absolute calibration in physical units. One way to gain confidence in the measurements is to compare the photometry of the stars other than Sirius measured by the MSX experiment. Table 4 shows the average deviation from the Cohen models for bands A–D, i.e., averaging the deviations for each star (we do not include Band E, 21.34  $\mu\text{m}$  in the following discussion because measurements in it have the largest errors and are also most susceptible to contamination from debris disk emission).

In general, the photometry of the bright stars in the various bands supports the view of high accuracy both for the CWW

**Table 4**  
Average Ratio to Sirius for MSX Bands A–D

| Band          | $\alpha$ Boo | $\alpha$ Tau | $\alpha$ Lyr <sup>a</sup> | $\beta$ Gem | $\gamma$ Dra | $\gamma$ Cru | $\beta$ Gem <sup>b</sup> |
|---------------|--------------|--------------|---------------------------|-------------|--------------|--------------|--------------------------|
| A             | 1.001        | 0.979        | 0.989                     | 0.980       | 0.974        | 0.981        | 0.994                    |
| B1            | 1.006        | 1.029        | 1.016                     | 0.979       | 1.012        | 1.040        | 1.000                    |
| B2            | 0.991        | 1.006        | 0.995                     | 0.949       | 1.001        | 1.007        | 0.970                    |
| C             | 0.991        | 0.990        | 1.048                     | 0.991       | 0.999        | 0.993        | 1.012                    |
| D             | 0.991        | 0.994        | 1.038                     | 0.991       | 0.998        | 0.977        | 1.010                    |
| E             | 0.984        | 0.996        | 1.172                     | 1.088       | 1.005        | 1.024        | 1.001                    |
| average (A–D) | 0.996        | 1.000        | 1.017                     | 0.978       | 0.997        | 1.000        | 0.997                    |

**Notes.**

<sup>a</sup> Value above 1 expected due to excess emission from debris disk.

<sup>b</sup> Using new spectral template computed for this paper.

(Cohen, Walker, & Witteborn) stellar templates and for the MSX measurements. Four of the stars deviate from expectations by  $<1\%$ .  $\alpha$  Lyr (=Vega) deviates in the direction of having an infrared excess, which is expected. However, the spectrum of  $\alpha$  Boo is of modest signal to noise, and that of  $\beta$  Gem shows a significant and unexplained deviation from the nominal ratios. We address the issues with these stars in Appendix A; primarily, we have replaced the relevant spectral segments with accurate Spitzer IRS spectra of similar stars, suitably normalized. For  $\beta$  Gem, we also make use of critically reviewed photometry from the literature. We find that the  $\alpha$  Boo spectrum is adequate for its use in the calibration calculations. We derive an improved template for  $\beta$  Gem. This template provides independent predictions for the MSX measurements for  $\beta$  Gem, which we ratio over the appropriate photometric bands to the observed ratio (i.e., similar to Table 4 of Price et al. 2004a). The individual values and average of these ratios are entered into Table 4. The upgraded template has completely removed the discrepancy between the measurements of  $\beta$  Gem and the other MSX targets, and it is now just 0.003 mag low. If we remove the somewhat discordant ratio for Band B2, the average is 0.004 mag high. That is, the behavior of  $\beta$  Gem confirms the MSX photometry of Sirius relative to it and the other stars to within 0.005 mag.

### 3.2.2. The MSX Reference-sphere Calibration

We now discuss the second part of the MSX results, the calibration obtained by observing reference spheres. This effort was strongly focused on demonstrating that the spectral templates from Cohen and collaborators, e.g., Cohen et al. (1995) and Cohen et al. (1996), were accurate (S. Price, private communication). This emphasis results in some challenges in applying them to alternative, and hopefully improved, calibrations. Our approach is discussed below.

The results of the observations of the calibrated reference spheres are described in Price et al. (2004a) and in more detail in Price et al. (2004b), quoted as biases relative to the Cohen et al. (1995) calibrated spectrum of Sirius. However, this process included making adjustments to the biases to match what was thought to be a robust defining photometric system based on Vega. They found that the flux difference between the Cohen et al. (1995) models of Vega and Sirius was about 1% more than those that were measured in the shorter-wavelength bands (free from the debris disk emission).<sup>14</sup> We might now attribute this effect to the as-yet-undiscovered hot excess revealed by interferometric

measurements of Vega (Absil et al. 2006, 2008). To reconcile the reference-sphere measurements with the models of Vega and Sirius, Price et al. (2004a) adjusted the reference-sphere biases by 1%. In effect, because they did not know about the hot excess of Vega, they made an ad hoc adjustment in the reference-sphere data. Price et al. (2004a) pointed out that the measured fluxes for their secondary standards are also offset by 1% in Bands A and B2. The improved reference spectrum for  $\beta$  Gem derived in Appendix A removes this offset for it, and the fact that many of the stars are offset by  $\sim 1\%$  in the opposite direction for Band B1 further calls it into question. We conclude that the difference between Vega and Sirius reflects primarily the hot excess of Vega plus measurement uncertainties rather than template errors or biases.

However, the validity of this adjustment can better be checked by returning to the MSX data as described in Price et al. (2004b). To be confident in our interpretation, we have thoroughly reviewed the results and independently derived the critical quantities for our calibration. We confirmed the results as described in Price et al. (2004a) with the following exceptions: (1) as discussed below, we apply the offsets in Table 9 of Price et al. (2004a) after removing the 1% increase discussed in that paper; and (2) we identified an error in the bias for band E, which should be  $-1.8\%$  rather than  $-2.5\%$ , or after removal of the 1%,  $-2.8\%$ . That is, to bring the CWW fluxes into agreement with the reference-sphere calibration, we have multiplied them by 0.996, 1.014, 1.029, and 1.019, respectively, at bands A, C, D, and E.

To derive a calibration, we apply the biases as just described to the in-band fluxes for MSX bands A, C, D, and E as they are determined from the CWW Sirius spectrum. We assign errors to these fluxes of 2%, 1.4%, 1.4%, and 3%, respectively, following the discussion in Price et al. (2004a) and Price et al. (2004b). We then adjust the normalization of the CWW Sirius spectrum to optimize the fit (minimum  $\chi^2$ ) to these measurements, finding it must be increased by 0.59% on top of the 1% adjustment discussed in the previous paragraph. The renormalized CWW spectrum then gives an estimate of the flux in the  $K_S$  band. However, the CALSPEC Sirius spectrum when normalized to the CWW one in the 7–20  $\mu\text{m}$  range is 0.7% fainter at 2.2  $\mu\text{m}$ ,<sup>15</sup> so this correction is applied to obtain a final value at 2.1603  $\mu\text{m}$  that is correctly referenced to the CALSPEC spectrum.

Price et al. (2004a) suggested that an error of 1.4% should be assigned to each of the six MSX spectral bands. Although it

<sup>14</sup> These models were, of course, the CWW models of the time.

<sup>15</sup> That is, the slope of the spectrum differs from the CALSPEC one by  $\sim 0.6\%$ – $0.7\%$  between 2 and 10  $\mu\text{m}$ , so this adjustment at 2.2  $\mu\text{m}$  compensates for the 0.59% one required at longer wavelengths.

**Table 5**  
Parameters of Solar-like Calibration Stars

| Star         | $T$ (K)           | Error (K) | Radius ( $R_{\odot}$ ) | Error ( $R_{\odot}$ ) | $K_S$ (mag)         | Error | $K_{S,\odot}$ | Error |
|--------------|-------------------|-----------|------------------------|-----------------------|---------------------|-------|---------------|-------|
| $\alpha$ Cen | 5795 <sup>a</sup> | 6         | 1.2234 <sup>b</sup>    | 0.0053                | -1.537 <sup>c</sup> | 0.008 | -28.299       | 0.013 |
| 18 Sco       | 5799 <sup>d</sup> | 3.8       | 1.010 <sup>e</sup>     | 0.009                 | 4.004 <sup>f</sup>  | 0.01  | -28.319       | 0.023 |
| 16 Cyg A     | 5803 <sup>d</sup> | 3.1       | 1.243 <sup>g</sup>     | 0.008                 | 4.431 <sup>h</sup>  | 0.01  | -28.292       | 0.018 |
| 16 Cyg B     | 5752 <sup>d</sup> | 3.5       | 1.127 <sup>g</sup>     | 0.007                 | 4.660 <sup>h</sup>  | 0.01  | -28.289       | 0.018 |

**Notes.**

<sup>a</sup> From Table 6.

<sup>b</sup> From Kervella et al. (2017).

<sup>c</sup> See Appendix B.

<sup>d</sup> From Kovtyukh et al. (2003).

<sup>e</sup> From Bazot et al. (2011).

<sup>f</sup> From Casagrande et al. (2014), transformed as in Koen et al. (2007).

<sup>g</sup> From Metcalfe et al. (2012).

<sup>h</sup> From Kidger & Martín-Luis (2003) transformed as in Rieke et al. (2022), plus 2MASS at half weight.

might be expected that some of the errors will be random and thus that averages will have a lower net error, this assumption does not allow for the possibility of overall systematic errors. An example would be an error in the measurement of the diameters of the reference spheres, which would affect all of the absolute calibrations by the same amount and in the same direction. There are also risks of systematic errors in the data itself and its analysis. Given the risk of systematic errors, we do not reduce the error from the combination of the multiple measurement bands. See Bohlin (2014) for a parallel argument arriving at the same conclusion. The flux for Sirius at  $2.1603 \mu\text{m}$  is  $15.20 \times 10^{-14} \text{ W cm}^{-2} \mu\text{m}^{-1}$ , entered along with a conservative error of 1.4% into Table 3.

### 3.3. Calibration via Solar-type Stars

Another route to an absolute calibration takes advantage of the presence of four nearby stars very similar to the Sun:  $\alpha$  Cen A, 18 Sco, and 16 Cyg A & B. With only small corrections for their differing diameters and slightly different temperatures, photometry of these stars can be compared directly with absolute flux measurements of the Sun to obtain a calibration of stellar infrared photometry. The approach we will use has some resemblance to the solar analog method originally used by Johnson, but in this case we base our calibration purely on infrared measurements to remove any dependency on optical data and optical-to-infrared colors. We will independently derive a pure solar analog calibration in Section 3.7.

The key parameters for these stars are summarized in Table 5. For comparison, the IAU recommended  $T_{\text{eff}}$  for the Sun is 5772 K (Prsa et al. 2016); the limb darkening for  $\alpha$  Cen A is also very similar to that of the Sun (Kervella et al. 2017), as one would expect from their general similarity. The largest difference between these stars and the Sun is their diameters. We assume that their outputs at  $K_S$  go as the radius of the star squared and as the ratio of their temperatures to that of the Sun. To justify the latter relation, we have shown over a much larger temperature range that a Rayleigh–Jeans approximation gives a reasonably accurate representation of the infrared output dependence on temperature (Su et al. 2022); see also Appendix C. The distance modulus to the Sun is  $-31.5721$  (from the definition of the astronomical unit), and to  $\alpha$  Cen A is  $-4.3671$  (from its parallax of  $747.17 \pm 0.61$  mas Kervella et al. 2016); the distance moduli of the other stars have been taken from Gaia. Interferometric searches for a hot excess have not been conducted for  $\alpha$  Cen A

or 18 Sco, but the average excess for 12 G-type stars from Absil et al. (2013) and Nunez et al. (2017) is only  $0.5 \pm 0.08\%$ , so we ignore this possibility in these cases. Absil et al. (2013) found no hot excess emission around 16 Cyg A or B, showing their  $K$ -band emission to be purely photospheric. The resulting estimates for the  $K_S$  magnitude of the Sun are entered into Table 5. The errors are the root square sum (RSS) combination of the errors in solid angle subtended by the star, and in its magnitude, parallax, and temperature.

All four results are closely consistent; we combine them by weighted average, with a final value for the  $K_S$  magnitude of the Sun as  $-28.298 \pm 0.009$ . We put this magnitude into physical units using the results from Gueymard (2018) and Coddington et al. (2021), both of which present accurately calibrated composite spectra of the Sun. Both spectra agree very closely in total flux across the  $K$ -band atmospheric window. However, in the  $H$ -band window, the Coddington et al. (2021) spectrum is on average 1.5% fainter.<sup>16</sup> The agreement with the colors of solar-type stars (see footnote) suggests that the Coddington et al. (2021) spectrum is the more accurate over the full infrared but does not call into question the agreement in the  $K$  band. Given the close agreement there, we average the two values.

To determine the flux from the Sun, we interpolate over the Br  $\gamma$  absorption (after masking points affected by the absorption) since it coincides with the  $K_S$  pivot wavelength of  $2.1603 \mu\text{m}$ . To do so, we fit a power law to the spectrum from  $2.00$ – $2.31 \mu\text{m}$ . Corrected from the magnitude of the Sun to zero magnitude, the flux at this wavelength is

<sup>16</sup> To test the cause of this discrepancy, we use the photometric colors of the Sun from Casagrande et al. (2012). These values are in the 2MASS photometric system, which has small offsets from an A0V-star zero magnitude system, as quantified by Rieke et al. (2008) and Roc Cutri, also in Rieke et al. (2008), and Maíz Appellániz & González (2018). We average the three determinations, giving double weight to the latter reference and derive corrections of 0.021 mag bluer in  $H - K_S$  and 0.015 mag redder in  $J - K_S$ . The original values of  $H - K_S = 0.076$  and  $J - K_S = 0.362$  (Casagrande et al. 2012) therefore become  $H - K_S = 0.055$  and  $J - K_S = 0.377$ , respectively, for a photometric system with zero color differences for an idealized A0V star. The agreement with tabulated colors of G2V dwarf stars is excellent, i.e.,  $H - K = 0.05$  and  $J - K = 0.37$  from Bessell & Brett (1988) and Tokunaga (1999). These values have been compared with the colors for the Gueymard (2018) Sun through synthetic photometry relative to a zero-point represented by the CALSPEC model of Sirius (Bohlin et al. 2017, 2020). The synthetic photometry returns  $H - K_S = 0.039$  and  $J - K_S = 0.381$ , i.e., the Gueymard (2018)-based color differences are 0.016 mag smaller than expected at  $H - K_S$  and 0.004 mag larger at  $J - K_S$ . Similar synthetic photometry for the Coddington et al. (2021) spectrum agrees very closely with the expected colors.



**Table 6**  
Temperature of  $\alpha$  Cen

| Temperature (K) | Error (K) | Source                                 |
|-----------------|-----------|----------------------------------------|
| 5771            | 23        | Ramírez & Meléndez (2005)              |
| 5792            | 16        | Heiter et al. (2015)                   |
| 5816            | 39        | Heiter et al. (2015) <sup>a</sup>      |
| 5847            | 27        | Porto de Mello et al. (2008)           |
| 5799            | 11        | weighted average of above <sup>b</sup> |
| 5793            | 7         | Boyajian et al. (2013)                 |
| 5795            | 6         | average of all                         |

**Notes.**

<sup>a</sup> Summary of other measurements.

<sup>b</sup> That is, of spectroscopically determined temperatures.

$4.213 \times 10^{-14} \text{ W cm}^{-2} \mu\text{m}^{-1}$ . Gueymard (2018) estimated the error in the solar spectrum as 0.9%; Coddington et al. (2021) estimated an error of <0.3%. At these levels, the uncertainty in the solar irradiance has little effect on the result; the error is dominated by the uncertainty in the  $K_S$  magnitude and in the diameters of the comparison stars. The derived absolute calibration in flux density is corrected to the equivalent for Sirius at a magnitude of  $-1.395$  and entered into Table 3.

### 3.4. Near Infrared Ground-based Calibrations

A series of measurements of Vega calibrated against a local standard infrared source is reported in Selby et al. (1983), Blackwell et al. (1983), and Booth et al. (1989). These measurements used narrowband spectral filters to minimize photometric corrections and to center on spectral regions clear of atmospheric absorptions. Five of these measurements are between  $2.2$  and  $3.8 \mu\text{m}$  and have quoted errors of 3% or 4%. They can be combined using a nominal spectrum to adjust those near  $3.8 \mu\text{m}$  to those near  $2.2 \mu\text{m}$ . Given the small hot excess of Vega, it is not obvious how to combine these measurements at highest accuracy, but given their modest accuracy, any plausible method will not degrade the final error. We have used the CALSPEC Sirius spectrum for these corrections, and obtain an estimated  $K$ -band flux for Vega. The indicated error is 1.7%; we will adopt 2%. This result can be transferred to Sirius accurately using the relative brightness of Vega and Sirius derived in Rieke et al. (2022). We carry out this transfer (a difference of 1.37 mag) and then use the CALSPEC Sirius spectrum to place the result at  $2.1603 \mu\text{m}$ . Equivalently, the Vega results can just be adjusted by the 0.025 mag difference from Rieke et al. (2022) between the Vega/Sirius difference and the magnitude of Sirius. The result for Sirius is entered into Table 3.

### 3.5. Calibration with the Viking IRTM

Rieke et al. (1985) described a set of measurements in which  $\beta$  Gem was compared with Mars while Mars was observed with the Viking Infrared Thermal Mapper (IRTM) in a manner that permitted constructing the whole-disk flux as viewed from Earth. It was possible to match the  $11.1 \mu\text{m}$  filter in the ground-based photometer very closely to the spectral bandpass of the IRTM, so the errors are dominated by the uncertainty of the whole-body Mars flux. The net uncertainty of the measurement is 3.3%. With the accurate SED of  $\beta$  Gem derived in Appendix A, which is referred directly to Sirius, this measurement can be transferred to that star. The resulting

calibration, using the Sirius spectrum to convey it to  $2.1603 \mu\text{m}$ , is  $15.53 \pm 0.51 \times 10^{-14} \text{ W cm}^{-2} \mu\text{m}^{-1}$  (see Table 3).

### 3.6. Inverted Infrared Flux Method

Blackwell & Shallis (1977) proposed using infrared measurements on the Rayleigh–Jeans part of a stellar spectrum to deduce its angular diameter and hence to convert photometry of the star to absolute physical units. This IRFM has been used widely since. Here we invert the approach and apply it to the F-type stars for which we have obtained direct photometric transfers to Sirius (Su et al. 2022). As discussed in Appendix C, IRAC Band 1 has significant advantages for applying the IRFM at high accuracy.

The IRFM depends on four stellar parameters: (1) the bolometric flux,  $F_{\text{bol}}$ ; (2) the flux density at the chosen infrared wavelength,  $F(\lambda_0)$ ; (3) the angular diameter of the star,  $\theta$ ; and (4) the stellar surface flux  $\phi(T_{\text{eff}}, g, \text{Fe/H}, \lambda_0)$ , where  $T_{\text{eff}}$  is the effective temperature,  $g$  is the surface gravity, and  $\lambda_0$  is the chosen wavelength. Typically,  $\phi(T_{\text{eff}}, g, \text{Fe/H}, \lambda_0)$  is determined through theoretical models, which would inject an undesirable uncertainty into using the IRFM for very accurate results. However, ratioing Equations (3) and (2) from Blackwell et al. (1979), we obtain

$$\phi(T_{\text{eff}}, g, \text{Fe/H}, \lambda_0) = \frac{F(\lambda_0)}{F_{\text{bol}}} \sigma T_{\text{eff}}^4, \quad (1)$$

where  $\sigma$  is the Stefan–Boltzmann constant. The terms on the right-hand side of this equation are very accurately determined for the Sun. Appendix C shows how we have used this as a basis to determine an empirical surface flux for the F-stars. Given this parameter and values of  $T_{\text{eff}}$  (and assuming main-sequence values for  $g$  and  $\text{Fe/H}$ ), we predict the flux density from a star, namely:

$$F(\lambda_0) = \frac{\pi \theta^2}{4} \times \phi(T_{\text{eff}}, g, \text{Fe/H}, \lambda_0), \quad (2)$$

where as derived in Appendix C for IRAC Band 1 ( $3.55 \mu\text{m}$ ),

$$\phi(T_{\text{eff}}, g, \text{Fe/H}, \lambda_0) = \frac{T_{\text{eff}} \times 198 \text{ W cm}^{-2} \mu\text{m}^{-1}}{5772 \text{ K} \times F(T_{\text{eff}})} \quad (3)$$

with

$$F(T_{\text{eff}}) = 1.111 \times 10^{-16} T_{\text{eff}}^4 - 3.753 \times 10^{-12} T_{\text{eff}}^3 + 4.919 \times 10^{-8} T_{\text{eff}}^2 - 2.879 \times 10^{-4} T_{\text{eff}} + 1.6196. \quad (4)$$

The stars to which we apply this method are listed in Table 7. We use (1) the measured limb-darkened angular diameters of the stars,<sup>17</sup> and (2)  $T_{\text{eff}}$ , both as listed in Table 7. We apply them through Equation (2) to estimate a flux density and compare the result with the measured flux density to derive an absolute calibration. We take the magnitude of Sirius to be  $-1.395$ ; for the magnitude of a star relative to it, we use the direct transfer in Su et al. (2022) plus, at half weight, the values from heritage  $K$ -band measurements (Rieke et al. 2022).<sup>18</sup>

<sup>17</sup> In Section 4.4 we discuss further the connection between infrared fluxes and limb-darkened stellar diameters.

<sup>18</sup> The results are unchanged if we use only the direct transfer measurements.



**Table 7**  
Inverted IRFM Calibrations

| HR          | HD     | $T_{\text{eff}}$<br>K | Ref. <sup>a</sup> | $\theta_{\text{LD}}$ <sup>b</sup><br>mas | $\phi$<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | Predicted Flux<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | Measured Flux<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | Ratio |
|-------------|--------|-----------------------|-------------------|------------------------------------------|-----------------------------------------------|-------------------------------------------------------|------------------------------------------------------|-------|
| 937         | 19373  | 5963                  | 1                 | $1.246 \pm 0.008$                        | 203                                           | $5.90 \times 10^{-15}$                                | $5.785 \times 10^{-15}$                              | 1.019 |
| 1543        | 30652  | 6425                  | 2,3               | $1.526 \pm 0.004$                        | 222                                           | $9.52 \times 10^{-15}$                                | $9.700 \times 10^{-15}$                              | 0.982 |
| 2943        | 61421  | 6562                  | 2,3               | $5.396 \pm 0.021$                        | 225                                           | $1.235 \times 10^{-13}$                               | $1.243 \times 10^{-13}$                              | 0.994 |
| 4540        | 102870 | 6055                  | 1                 | $1.431 \pm 0.006$                        | 212                                           | $8.01 \times 10^{-15}$                                | $8.009 \times 10^{-15}$                              | 1.004 |
| 5404        | 126660 | 6220                  | 3                 | $1.109 \pm 0.007$                        | 213                                           | $4.85 \times 10^{-15}$                                | $4.89 \times 10^{-15}$                               | 0.993 |
| 5933        | 142860 | 6279                  | 2,3               | $1.217 \pm 0.005$                        | 216                                           | $5.91 \times 10^{-15}$                                | $5.84 \times 10^{-15}$                               | 1.012 |
| 7061        | 173667 | 6359                  | 2,3               | $1.000 \pm 0.006$                        | 216                                           | $4.04 \times 10^{-15}$                                | $3.97 \times 10^{-15}$                               | 0.996 |
| average     |        |                       |                   |                                          |                                               |                                                       |                                                      | 1.000 |
| rms scatter |        |                       |                   |                                          |                                               |                                                       |                                                      | 0.013 |

**Notes.**

<sup>a</sup> References: 1: Kovtyukh et al. (2003), nominal errors  $< 10$  K; 2: Malyuto & Shvelidze (2011), nominal errors 30–40K; 3: Muñoz Bermejo et al. (2013), nominal errors  $\sim 20$  K.

<sup>b</sup> Boyajian et al. (2012) except for HR 2943, where we average the values from Aufdenberg et al. (2005) and Chiavassa et al. (2012). We reduce all of the values quoted except those for HR 2943 by 0.5% to compensate for their use of 1D limb darkening models; the value for HR 2943 is based on 3D models.

Our procedure is to adjust the assumed calibration until the average of the ratios of predicted-to-measured flux densities for all of the stars is unity. For IRAC Band 1,  $\lambda_{\text{pivot}} = \lambda_0 = 3.55 \mu\text{m}$ , and we take the measured flux density to be

$$F(\lambda_0) = F_{\text{ZP}} \times 10^{(-0.4 \times \text{mag})} \text{W cm}^{-2} \mu\text{m}^{-1}, \quad (5)$$

where mag is the magnitude in IRAC Band 1 and  $F_{\text{ZP}}$  is the zero-point flux density, i.e., the calibration, that is adjusted to agree with the value derived from the angular diameter and temperature.

Because the M3.5 companion makes a low but not well-determined contribution to the photometry of HD 215648, we discard this star from the calculations. One of the remaining stars, HD 173667, has a significant excess detected in interferometry at  $2 \mu\text{m}$  (Núñez et al. 2017), and we subtract this from the photometry, assuming the same value relative to the stellar output in IRAC band 1. Two remaining ones, HD 61421 and HD 12660, do not have interferometric measurements to look for hot excesses; we make no corrections to their photometry. The remaining stars, HD 19373, HD 30652, HD 102870, and HD 142860, have no measured excess (Absil et al. 2013; Núñez et al. 2017), so no corrections are needed. The results are listed in Table 7. The rms scatter around unity is only 0.013. The nominal error is a combination of 0.3% uncertainty in the solar measurements (Gueymard 2018; Coddington et al. 2021) used to determine  $\phi$ , plus terms for the uncertainties in radii, which dominate, and stellar effective temperatures.

Nominally, the errors in this calibration combine to only 0.8%, but there is a possibility of larger systematic errors due to, for example, the choice of limb darkening model (1D or 3D stellar atmospheric modeling) in the interpretation of the interferometry (e.g., Karovicova et al. 2022). We use diameters determined in the near-infrared, where limb darkening is significantly smaller than in the visible where it is discussed by Karovicova et al. (2022). Whereas they found a difference in diameters from 1D and 3D models at the 1.1% level, scaling by the relative level of limb darkening suggests that the corresponding difference in the near-infrared would be only  $\sim 0.5\%$ . Aufdenberg et al. (2005) and Bigot et al. (2006) compared the systematic effects in interferometric angular diameters determined in the visible and infrared and concluded

that the sensitivity to systematic effects at the latter wavelengths is no more than 0.5%. A systematic shift in the diameters by 0.5% results in a shift in the derived calibration of 0.8%. We therefore: (1) reduce all of the diameters determined using 1D limb darkening behavior by 0.5%; and (2) adopt a final rms error value of 1.5% to allow for possible additional errors in the diameters. The resulting calibration and its error are entered into Table 3. For reference, if we had used the results for 1D limb darkening behavior, our overall calibration would become brighter by 0.13% once all of the other methods are folded in; i.e., the effect is well within the errors.

### 3.7. Solar Analog Calibration

The traditional solar analog method (Johnson 1965) can also be applied as an independent test of the calibrations just discussed. It is based on obtaining an accurate measure of the Sun in some standard photometric band on a scale consistent with stellar measurements in the same band. An absolutely calibrated solar spectrum can then be renormalized to predict the spectrum of stars of the same spectral type as the Sun (i.e., G2V). Since the calibration stars are unlikely to be identical to the Sun, adjustments are made to their colors on the basis of comparing effective temperatures. As a starting point for this approach, we take the  $V$  magnitude of the Sun to be  $-26.74 \pm 0.02$  (see Section 2.2).

The challenge in this approach is to determine an accurate solar  $V - K_S$  from solar-type stars by matching the effective temperatures accurately since the color is strongly dependent on temperature. Rieke et al. (2008) did this through conventional stellar temperature estimates, but Casagrande et al. (2012) improved the procedure by using temperature-sensitive line depth ratios (LDRs) to determine temperatures. Their color for the Sun is  $V - K_S = 1.560$ . To put this value on a photometric system with A0V dwarf stars defining zero color, we need to subtract 0.015 mag (Maíz Appellániz & González 2018), to obtain  $V - K_S = 1.545$  for the Sun. Taking the  $V$  magnitude just determined (Section 2.2), we find the  $K_S$  magnitude of the Sun to be  $-28.285 \pm 0.020$ . This agrees within the errors with the value of  $-28.298 \pm 0.009$  from direct transfer at  $K$  based on  $\alpha$  Cen A and the other solar-like stars (Section 3.3).

We estimate the error as follows. As already discussed, we take the error of the  $V$ -band magnitude of the Sun to be 2%.

The line-depth-ratio based  $V - K_S$  of the Sun has an estimated error of 1.3% (Casagrande et al. 2012). We assign an error of 0.4% to the correction of the 2MASS  $K_S$  mag to an A0V zero-based magnitude system (Maíz Appellániz & González 2018), and 0.3% to the solar flux (see Section 2.2). The net error is 2.5%.

### 3.8. A-Star Spectral Energy Distributions

Early A-type stars were originally selected as the zero-point of the stellar photometric system under the belief that their spectra could be modeled from first principles relatively accurately (e.g., Johnson & Morgan 1953; Dreiling & Bell 1980). This belief has been confounded by effects not included in the models, such as rapid rotation and the accompanying temperature gradients over their surfaces, and the excess emission due to circumstellar dust throughout the infrared. However, Sirius appears to retain all of the good features originally presumed in general, so we can use models of its SED to extrapolate from the optical calibration into the infrared.

We have already mentioned how we use the theoretical SED of Sirius to relate absolute calibrations at different wavelengths to each other and, for the sake of comparison, to convert them to equivalent calibrations at  $2.1603 \mu\text{m}$ . The same spectrum can itself be used as an absolute calibration by normalizing it at  $5557.5 \text{ \AA}$  to the calibration at this wavelength from Section 2. Previous error estimates for this extrapolation have been  $\sim 2\%$  (Bohlin et al. 2011; Bohlin 2014; Bohlin et al. 2017). Given progress in modeling the Sirius SED (Bohlin 2014; R. Bohlin, private communication) and the improved absolute calibration in the visible (this work; Bohlin et al. 2019), we set the error to 1.5%. Because demonstrating the consistency of the Sirius SED across the visible and infrared is one of our goals, although we could put the error smaller, we avoid doing so to be sure that Sirius itself has only a minor influence on the final calibration.

### 3.9. White Dwarf-based Calibration

White-dwarf stars are widely used for calibration in the ultraviolet and optical because they are well modeled and their SEDs are simple, without complexes of spectral features. They have also been used to extrapolate from the Megessier (1995) calibration into the infrared. Bohlin et al. (2011) tested this procedure with Spitzer IRAC photometry and found that the white dwarf extrapolated calibration was in agreement with the calibrations from A-type and G-type stars within the errors, although there was a  $\sim 2\%$ – $3\%$  offset in this comparison.

Here we update this result using additional photometry and improved models for white dwarf SEDs. We use the IRAC measurements of G191B2B, GD 71, and GD 153 (GD 749B was also measured but is too faint for a solid calibration). The photometry is described by Krick et al. (2021), but has been re-reduced to higher accuracy by Bohlin et al. (2022). For the primary stars we will use, both reductions are in very close agreement; e.g., the average change in flux for the three white dwarfs and for the two colors is increased by 0.67% and for the prime calibrator star HD 165459 by 0.68%. Bohlin et al. (2022) quote fluxes for the white dwarfs assuming the calibration derived by Carey et al. (2012). However, to derive an independent calibration, we determine the brightnesses of the WDs relative to Sirius rather than assuming a calibration. To do so, we used the 2MASS photometry for all of the stars with

fluxes from Bohlin et al. (2022) with IRAC Band 1 fluxes  $< 1 \text{ Jy}$  (to avoid linearity concerns). We checked this result with five stars with independent accurate measurements,<sup>19</sup> transformed to 2MASS and for them got on average perfect agreement. The average offset is then  $0.002 \pm 0.006$ . We apply this adjustment to the measurements of Bohlin et al. (2022) for all three WDs and for both IRAC bands 1 and 2.

There is also high-accuracy  $K$ -band photometry for two of the white dwarfs as part of the UKIRT calibration campaign (Hawarden et al. 2001; Leggett et al. 2006). We transform these measurements to  $K_S$  for comparison with Sirius.

All of the resulting photometry relative to Sirius can be found in Table 8. These measured values need to be compared with the models used to extrapolate the calibration from the visible to the infrared. Models for these stars described by Bohlin et al. (2020) were taken from CALSPEC. We carried out synthetic photometry on the three WDs and on Sirius to compare with the observations relative to Sirius just described. We show the results in Table 8. The average difference from the extrapolation via the spectrum of Sirius is  $0.002 \pm 0.005$  magnitudes fainter if the calibration had been obtained via the white-dwarf stars. That is, the agreement is virtually perfect. Nonetheless, if the error for the Sirius extrapolation is taken to be 1.5%, we assume an error of 2% for the white dwarf extrapolation, an increase that is supported by the variations in the differences of synthetic photometry versus observed delta magnitudes between Sirius and the WDs from one case to another in Table 8.

### 3.10. Diameter of Sirius

Bohlin et al. (2017) showed that their models of Sirius very nearly match the measured diameter of the star. The calibration of their models is based on a zero-point of  $3.44 \times 10^{-12} \text{ W cm}^{-2} \mu\text{m}^{-1}$  for Vega at  $0.55575 \mu\text{m}$ , or  $1.35 \times 10^{-11} \text{ W cm}^{-2} \mu\text{m}^{-1}$  for Sirius. The 2020 CALSPEC Sirius spectrum translates this to a zero-point of  $4.28 \times 10^{-14} \text{ W cm}^{-2} \mu\text{m}^{-1}$  at  $2.16 \mu\text{m}$ , i.e., 0.13% higher than our calibration. Adjusting to our value would make the agreement even better, within the  $1\sigma$  error bar of the diameter measurement.

### 3.11. Summary

We have adopted Sirius as the fiducial star for absolute calibration in the visible through the mid-infrared. The traditional fiducial calibration star in the visible is Vega, so we have updated measurements of it and then transferred the result to Sirius through the relative magnitudes of the two stars. The results are reported in Table 1. This summary includes the inputs to the calibration by Megessier (1995), which have provided the basis for the calibration for the past 27 yr. We supplemented these measurements with additional absolutely calibrated observations of additional stars, referred to Vega. We have also used the  $V$  magnitude of the Sun to provide an additional calibration. All of these approaches are in good agreement, as summarized in Table 1.

In the infrared, we report measurements directly traced to Sirius in Table 3. The measurements are at a range of infrared wavelengths; we trace them to equivalent calibrations of the flux density from Sirius at  $2.1603 \mu\text{m}$  through

<sup>19</sup> HD 2811, Carter (1990); HD 101452, Carter (1990), McGregor (1991); HD 116405, Kidger & Martín-Luis (2003); HD 158485, Kidger & Martín-Luis (2003); and GSPC P330-E, Persson et al. (1998).

**Table 8**  
White Dwarf Synthetic Photometry Results

| Parameter                         | Sirius                     | GD191B2B                  | GD71                                        | GD153                    |
|-----------------------------------|----------------------------|---------------------------|---------------------------------------------|--------------------------|
| Band                              |                            | $K_S$                     | $\lambda_{\text{piv}} = 2.1603 \mu\text{m}$ |                          |
| $C_{\text{fit}}^a$                | $2.801706 \times 10^{-12}$ | $6.985 \times 10^{-18}$   | $2.383 \times 10^{-18}$                     | $1.712 \times 10^{-18}$  |
| $\alpha_{\text{fit}}^a$           | $-3.77731$                 | $-3.94304$                | $-3.95744$                                  | $-3.94387$               |
| Flux ( $\lambda_{\text{piv}}^a$ ) | $1.527 \times 10^{-13}$    | $3.351 \times 10^{-19}$   | $1.131 \times 10^{-19}$                     | $8.21 \times 10^{-20}$   |
| Predicted $\Delta$ mag            |                            |                           | 15.326                                      | 15.674                   |
| Observed $\Delta$ mag             |                            |                           | 15.308 <sup>b</sup>                         | 15.664 <sup>b</sup>      |
| Band                              |                            | IRAC Band 1               | $\lambda_{\text{piv}} = 3.550 \mu\text{m}$  |                          |
| $C_{\text{fit}}^a$                | $3.13291 \times 10^{-12}$  | $6.99645 \times 10^{-18}$ | $2.419 \times 10^{-18}$                     | $1.720 \times 10^{-18}$  |
| $\alpha_{\text{fit}}^a$           | $-3.86961$                 | $-3.944182$               | $-3.97145$                                  | $-3.95004$               |
| Flux ( $\lambda_{\text{piv}}^a$ ) | $2.3269 \times 10^{-14}$   | $4.728 \times 10^{-20}$   | $1.579 \times 10^{-20}$                     | $1.154 \times 10^{-20}$  |
| Predicted $\Delta$ mag            |                            | 14.230                    | 15.421                                      | 15.762                   |
| Observed $\Delta$ mag             |                            | 14.243                    | 15.428                                      | 15.778                   |
| Band                              |                            | IRAC Band 2               | $\lambda_{\text{piv}} = 4.494 \mu\text{m}$  |                          |
| $C_{\text{fit}}^a$                | $3.00276 \times 10^{-12}$  | $6.9524 \times 10^{-18}$  | $2.3889 \times 10^{-18}$                    | $1.7148 \times 10^{-18}$ |
| $\alpha_{\text{fit}}^a$           | $-3.84086$                 | $-3.93983$                | $-3.96106$                                  | $-3.94611$               |
| Flux ( $\lambda_{\text{piv}}^a$ ) | $9.351 \times 10^{-15}$    | $1.8658 \times 10^{-19}$  | $6.2098 \times 10^{-20}$                    | $4.5588 \times 10^{-20}$ |
| Predicted $\Delta$ mag            |                            | 14.250                    | 15.444                                      | 15.780                   |
| Observed $\Delta$ mag             |                            | 14.260                    | 15.433                                      | 15.803                   |

**Notes.**

<sup>a</sup> These values are for the power-law fit to the stellar continuum (the normalizing constant  $C$  and the slope  $\alpha$ ) and the resulting value for the flux density at the pivot wavelength, in units of  $\text{W cm}^{-2} \mu\text{m}^{-1}$ .

<sup>b</sup> From Hawarden et al. (2001), Leggett et al. (2006).

normalizing the SED of Sirius to the measured value and using it to determine the equivalent flux density at  $2.1603 \mu\text{m}$ . Although they use a variety of approaches, all eight procedures considered yield consistent values, as shown in Table 3. In addition, the visual and infrared calibrations are in excellent agreement, yielding a consistent calibration across the entire visual to  $30 \mu\text{m}$  range.

#### 4. Discussion

Based on the full suite of determinations in Tables 1 and 3, the nominal uncertainties in our final calibrations in both the visual and infrared are only  $\sim 0.6\%$ . In this section, we use jackknife analyses to test for systematic errors that might indicate significantly larger uncertainties. We find that any such errors are small. Then we discuss the zero-point of the stellar magnitude system, in light of our suggestion that it be defined by the SED of Sirius. There is a significant difference between the traditional zero-point and the color difference  $V - K_S$  for Sirius. We attribute this offset to anomalies in the SEDs of normal A-stars associated with both rapid rotation and near-infrared excesses caused by hot dust. This discussion leads to an evaluation of the suitability of the three suggested types of secondary calibration stars—A-stars, G-stars, and white dwarfs—for use at the accuracy level now available for the absolute calibration. We finally provide a short demonstration of the ability of the proposed calibration, along with the formalism we have developed to determine stellar emissivity, to return very accurate stellar diameters through use of the IRFM.

##### 4.1. Jackknife Analysis

The visible and infrared calibrations are based on a number of approaches, each of which can be subject to systematic

**Table 9**  
Visible Jackknife Tests for Vega

| Excluded Method            | Flux Density<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | Error<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | Offset<br>Percent | Error<br>Percent |
|----------------------------|-----------------------------------------------------|----------------------------------------------|-------------------|------------------|
| none                       | $3.473 \times 10^{-12}$                             | $0.018 \times 10^{-12}$                      | ...               | 0.51             |
| Hayes                      | $3.475 \times 10^{-12}$                             | $0.018 \times 10^{-12}$                      | 0.1               | 0.53             |
| Tüg                        | $3.457 \times 10^{-12}$                             | $0.027 \times 10^{-12}$                      | −0.5              | 0.77             |
| Terez                      | $3.479 \times 10^{-12}$                             | $0.020 \times 10^{-12}$                      | 0.17              | 0.59             |
| Willstrop                  | $3.474 \times 10^{-12}$                             | $0.018 \times 10^{-12}$                      | 0.02              | 0.53             |
| Sun                        | $3.474 \times 10^{-12}$                             | $0.018 \times 10^{-12}$                      | 0.02              | 0.53             |
| all but Hayes <sup>a</sup> | $3.45 \times 10^{-12}$                              | $0.069 \times 10^{-12}$                      | −0.7              | 2.0              |
| all but Tüg                | $3.49 \times 10^{-12}$                              | $0.024 \times 10^{-12}$                      | 0.37              | 0.68             |
| all but Terez              | $3.46 \times 10^{-12}$                              | $0.036 \times 10^{-13}$                      | −0.5              | 1.0              |
| all but Willstrop          | $3.46 \times 10^{-12}$                              | $0.069 \times 10^{-13}$                      | −0.3              | 2.0              |
| all but Sun                | $3.465 \times 10^{-12}$                             | $0.07 \times 10^{-12}$                       | −0.2              | 2.0              |

**Note.**

<sup>a</sup> The purpose of these entries is to look for systematic errors in all of the measurements by a given group or author. Since there is only one such value for Hayes, we just repeat it. However, for Tüg, the entry is the weighted average of Tüg et al. (1977, 1977) via 109 Vir, and Tüg (1980a), and for Terez, it combines Terez & Terez (1979) and Terez (1977); see Table 1.

errors that are not captured in the weighted averages we have used to generate final calibration flux densities. To test for this possibility, we have conducted jackknife tests in which we systematically eliminate one or more of the calibration approaches and recompute the weighted average to see if it has changed outside of expectations. Where a single investigator or group carried out a number of the calibrations we have used, in the jackknife tests we have eliminated all of them, since it is likely that systematic errors would be common to



**Table 10**  
Infrared Jackknife Tests for Sirius

| Excluded Method        | Flux Density<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | Error<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | Offset<br>Percent | Error<br>Percent |
|------------------------|-----------------------------------------------------|----------------------------------------------|-------------------|------------------|
| none                   | $15.258 \times 10^{-14}$                            | $0.090 \times 10^{-14}$                      | ...               | 0.59             |
| MSX                    | $15.269 \times 10^{-14}$                            | $0.100 \times 10^{-14}$                      | 0.08              | 0.65             |
| Solar                  | $15.266 \times 10^{-14}$                            | $0.104 \times 10^{-14}$                      | 0.06              | 0.68             |
| Near Infrared          | $15.223 \times 10^{-14}$                            | $0.094 \times 10^{-14}$                      | -0.22             | 0.62             |
| Viking IRTM            | $15.249 \times 10^{-14}$                            | $0.092 \times 10^{-14}$                      | -0.06             | 0.64             |
| Inverted IRFM          | $15.275 \times 10^{-14}$                            | $0.098 \times 10^{-14}$                      | 0.32              | 0.64             |
| Solar Analog           | $15.245 \times 10^{-14}$                            | $0.093 \times 10^{-14}$                      | -0.03             | 0.61             |
| Sirius SED             | $15.236 \times 10^{-14}$                            | $0.098 \times 10^{-14}$                      | -0.08             | 0.64             |
| White Dwarf            | $15.246 \times 10^{-14}$                            | $0.094 \times 10^{-14}$                      | -0.02             | 0.62             |
| Sun                    | $15.290 \times 10^{-14}$                            | $0.108 \times 10^{-14}$                      | 0.03              | 0.80             |
| Sirius & WD            | $15.229 \times 10^{-14}$                            | $0.104 \times 10^{-14}$                      | -0.12             | 0.68             |
| all but Sirius<br>& WD | $15.316 \times 10^{-14}$                            | $0.184 \times 10^{-14}$                      | 0.38              | 1.2              |

them all (e.g., if there is an error in the calibration of a reference source). The results, both for the visible and infrared, are shown in Tables 9 and 10. They are very encouraging that there are no major systematic errors.

#### 4.2. The Magnitude System

A-type stars were selected to define the zero-point of the photographic observations of the North Polar Sequence (NPS) by Leavitt (1917), and adopted as a uniform zero-point to reconcile a broad range of visual and photographic photometry in the first general assembly of the IAU (Seares et al. 1922). Photoelectric measurements of the NPS stars were obtained by Stebbins et al. (1950), and soon after Johnson & Morgan (1953) codified A0V stars as the zero-point of the photoelectric photometric system. With time, Vega became the premier zero-point star, in part because it was bright enough for accurate measurements even with early instruments in the mid-infrared. The adoption of A0V stars and particularly Vega to define the zero-point was therefore not the result of an extensive trade study, nor did it fold in modern information about the nature of these stars.

An indication of a problem with this definition of the zero-point was the discovery that Vega is both a rapidly rotating star with a substantial temperature gradient from pole to equator (Aufdenberg et al. 2006) and also has a subtle excess at  $2 \mu\text{m}$  and longer wavelengths, probably due to hot dust (Absil et al. 2006). The effects of these phenomena on the SED of the star are not amenable to accurate a priori determination, as is necessary for a zero-point definition. They also push the colors of the star to be redder than that of a pure A0V star: Rieke et al. (2008) estimated the  $K_S$  magnitude to be  $-0.036$ , which, with the  $V$  magnitude of  $+0.033$  (Mermilliod 1991), results in  $V - K_S = 0.069$ . There has been a tendency to dismiss Vega as an anomalous case and to continue as if other A0V stars are much better behaved. However, comparing the  $V$ -band and  $K_S$ -band magnitudes derived here and from the literature implies that the  $V - K_S$  of Sirius is  $\sim -0.04$  to  $-0.05$ , an equally worrisome result since this makes it bluer than the traditional zero-point by almost as much as Vega is redder. We discuss these issues below in Section 4.3.1, where we suggest they arise from a general unsuitability of normal A-stars to be used for an accurate zero-point. We demonstrate there that

Sirius can define an improved zero-point while retaining the A-star foundation.

#### 4.3. Secondary Calibrator Stars for Consistent Visible to Infrared Calibration

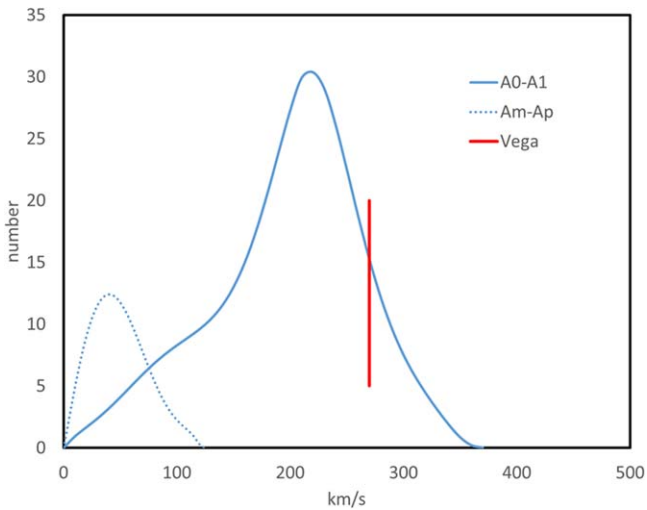
The focus of this paper is to demonstrate a consistent and accurate absolute calibration on very bright stars, primarily Sirius. Application of this calibration depends on selecting secondary standard stars that are much fainter and distributed widely over the sky, hence in general less well studied. In this section we discuss the issues that may be encountered with the common types of secondary standards: A-dwarf stars, G-dwarf stars, and white dwarfs (Gordon et al. 2022). The selection of these types is based on our ability to model their outputs at high fidelity. Accurate models allow interpolating and extrapolating the calibration over the full range of wavelengths desired.

In addition, any standard star must be free of any variability at a level that is significant compared with the accuracy of the desired calibration. The Kepler, CoRoT, and TESS satellites among others have alerted us to how ubiquitous low level variations are among stars, frequently at levels that compromise their use as secondary standards (e.g., Mullally et al. 2022). In general, preserving the calibration accuracy requires stars whose output is stable to  $\sim 0.2\%$ – $0.3\%$ , i.e., 2–3 mmag. Approximately 92% of G-dwarf (and  $\sim 90\%$  of F-dwarf) stars satisfy this requirement, and  $\sim 99\%$  are constant to within 1% (Ciardi et al. 2011), 97% of white dwarfs are constant to within 1% (Hermes et al. 2017), and  $\sim 96\%$  of A-dwarf stars are constant to 1% but only  $\sim 75\%$  to  $0.2\%$ – $0.3\%$  (Ciardi et al. 2011). Unless stability can be determined directly, A-stars are at higher risk for very demanding calibration. Assuming that variability has been ruled out to the necessary level, we discuss each of the three types in turn.

##### 4.3.1. A Dwarfs

We now explore the causes of the offsets in magnitudes for A-stars as discussed in Section 4.2. We find that they undermine the use of normal A0V stars to define photometric zero-points to the accuracy of the current absolute calibration.

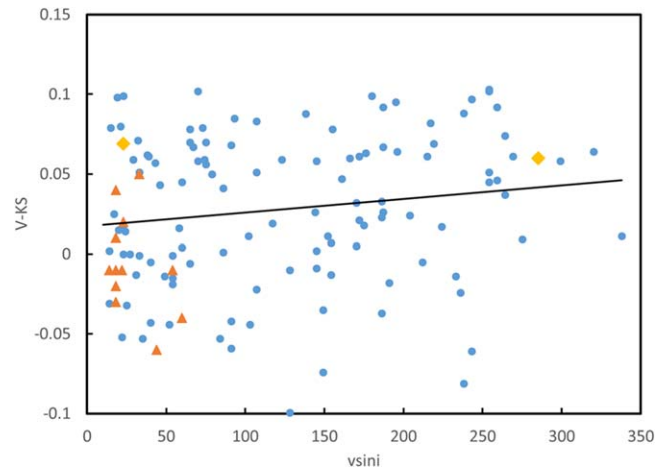
One contributor to the offset is that most normal A0V stars rotate rapidly and hence have significant temperature gradients from their poles to their equators, just as is the case for Vega. Any star of  $\sim 9000$  K with a significant temperature gradient will appear hotter in the visible range than in the infrared, and thus will have a “cooler” visible-to-infrared SED than implied by its spectral classification in the visible. We show the distribution of equatorial rotation velocities for A0–A1V stars from Abt & Morrell (1995) in Figure 2. We have marked the equatorial rotation rate for Vega (Aufdenberg et al. 2006); although high, it is by no means out of family. Thus, many other A0V stars are expected to have thermal gradients on scales up to that on Vega and to exhibit red  $V - K_S$  colors. Figure 2 also shows that Am and Ap stars, with compositional peculiarities, have as a group much slower rotation; Sirius is classified as A0mA1Va (Gray et al. 2003), consistent with its slow rotation (Takeda 2020). Unfortunately, despite their slower rotation, Am stars can also be significantly variable (Balona et al. 2015), so they cannot all be used as standards.



**Figure 2.** Equatorial rotation velocities for early A-stars from Abt & Morrell (1995), along with the equatorial rotation velocity value for Vega from Aufdenberg et al. (2006).

Figure 3 expands on these points.<sup>20</sup> It shows the behavior of A0V stars in  $V - K_S$  as a function of  $v \sin i$ , compared with similar information for Am and Ap stars. The latter are clustered around  $V - K_S = 0$  (the average is  $-0.005 \pm 0.008$ ).<sup>21</sup> The normal A-stars are significantly offset toward redder  $V - K_S$  even at low  $v \sin i$ , and the trend is toward a larger offset as the  $v \sin i$  increases. The difference in the average  $V - K_S$  for the normal stars versus the Am and Ap ones is  $0.03 \pm 0.01$ . With the additional effect of hot excesses discussed in the next paragraph, this is sufficient to explain the offset of the Sirius  $V - K_S$  from the traditional A0V star zero color. To emphasize that the  $v \sin i$  value is degenerate between the equatorial velocity and the stellar inclination, in Figure 3 we also plot Vega (high velocity small inclination) and 109 Vir (large inclination, probably similar equatorial velocity).

Many A stars also have hot excesses detected through interferometry, most often at  $H$  and  $K$  (e.g., Absil et al. 2013) but also at  $10 \mu\text{m}$  (e.g., Ertel et al. 2020, and references therein). These excesses are obviously not included in the stellar models used for calibration and hence bias the observations above the predictions of the models. We have estimated the effect of these excesses by computing the weighted average of the percentage excess above the photospheres in the  $K$  band, using the data from Absil et al. (2013) and N  nez et al. (2017). This sample consists of 23 A-type stars, and the average excess for the detected cases is  $0.99\% \pm 0.07\%$ . To account for the upper limits, we used the Kaplan-Meier test and found a possible range of 1%–1.5%, depending on how the nondetections are treated. That is, the average  $V - K_S$  of a large sample of A stars will be high compared with models by this amount, even if the effects of rotation are taken into account. Individual stars show hot



**Figure 3.** Dependence of  $V - K_S$  on  $v \sin i$  for A0V stars, using photometry from 2MASS and Hipparcos. The blue circles are “normal” stars while the red triangles are Am(0)V and Ap(0)V stars; the line is a fit to the normal stars. The orange diamonds are for Vega (small  $v \sin i$ ) and 109 Vir; because they are bright and saturated in recent photometry, the photometry is from Leggett et al. (1986), Rieke et al. (2008), and Mermilliod (1991). The  $v \sin i$  for the latter star (Royer et al. 2007) is virtually identical to the equatorial velocity for Vega (Aufdenberg et al. 2006). The scatter in  $V - K_S$  substantially exceeds that expected from the photometric errors ( $\leq 0.03$  mag). The underlying cause for the scatter is illustrated by the comparison of the Am and Ap stars with Vega, all with similar values of  $v \sin i$  but significantly different in  $V - K_S$ .

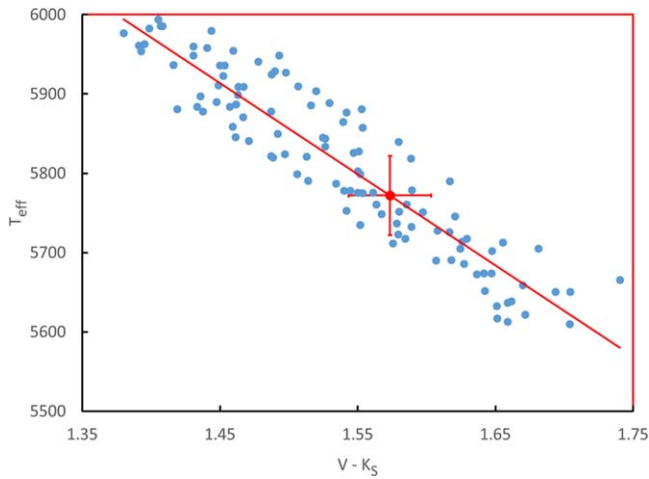
excesses up to 3% or even higher (N  nez et al. 2017). Unlike the equatorial rotation, which can in principle be estimated observationally (e.g., comparing rates from variability due to starspots with  $v \sin i$ ), the only way to identify these hot excesses is interferometry, which is impractical on faint calibrator stars. There can also be excesses due to debris disks; 45%–25% of A-stars between the ages of 10 and 500 Myr have detectable excesses, respectively, at  $24 \mu\text{m}$  (G  sp  r et al. 2013), and often these excesses persist down to wavelengths of  $\sim 10 \mu\text{m}$  (Ballering et al. 2014; Chen et al. 2014).

In summary, despite a history longer than a century for use as calibrator stars, normal A-stars pose a number of issues for use at very accurate levels across a broad spectral range. The typical rapid rotation can result in significant color trends, and the effect is difficult to detect because spectra show only the rotational velocity projected along the line of sight and are degenerate with regard to equatorial velocity. In addition, the excesses due to hot dust can contribute 1%–3% additional flux in the near- and mid-infrared that cannot be predicted or detected except through interferometry. These issues result in needing four free parameters to describe the SED of a normal A-star: (1) temperature; (2) equatorial rotation rate; (3) inclination; and (4) amount of hot near-infrared excess. As a result, the SED cannot be described simply from first principles. Finally, compared with other types of calibration stars, A-stars have a higher incidence of variability at levels that can undermine their use.

Some of these issues can be mitigated by using Am or Ap stars, because they are slow rotators. Whether they also have a reduced incidence of hot near-infrared excesses is not known. A-stars of all types do have the advantages of being distributed widely on the sky, with a large dynamic range in brightness, and having smooth continua between the prominent hydrogen line absorptions. They are useful over modest spectral ranges, even if there are problems in using them across broad ones. In particular, they provide smooth SEDs predictable to within the

<sup>20</sup> The data for the figure are from 2MASS and Hipparcos, with  $H_p$  translated to  $V$  according to Harmanec (1998). There are a significant number of outliers due to stellar multiplicity, misclassifications, and other effects, so we have trimmed cases with  $-0.1 < V - K_S < 0.1$ ; if anything, this trimming limits the apparent effect of rotation on increasing this color difference.

<sup>21</sup> That this value lies so close to zero is somewhat coincidental, since it relies on the definition of the 2MASS magnitudes, which was based on normal A stars.



**Figure 4.**  $V - K_S$  vs.  $T_{\text{eff}}$  for dwarf stars with temperatures similar to that of the Sun. The temperatures have been taken from the line depth ratio (LDR) results of Kovtyukh et al. (2003), the  $K_S$  photometry from 2MASS or heritage photometry transformed to 2MASS, and the  $V$  photometry from Hipparcos, transformed from  $H_p$  to  $V$ , as in Harmanec (1998). The central red point is at the solar temperature (5772 K) with error bars of  $\pm 0.03$  in  $V - K_S$  and  $\pm 50$  K in  $T_{\text{eff}}$ . The line is the linear fit given in Equation (6).

issues of the hot dust excesses across the infrared, where the temperature gradients have only modest effects.

#### 4.3.2. Solar-type Stars

Johnson (1965) introduced the use of the absolute measurements of the Sun in comparison with solar-type stars as a method of absolute calibration. The appeal of the approach is that the actual spectrum of the Sun is used rather than a theoretical model for some other type of calibrator star. This method has been used in a number of subsequent studies (e.g., Wamsteker 1981; Campins et al. 1985; Rieke et al. 2008). However, it has fallen into bad repute (Price 2004; Bohlin et al. 2014). The first reason is that the stars are cooler than A-stars and hence the optical to near-infrared colors are a steep function of the stellar temperature, which must be determined accurately to get good results. This problem has been dealt with in Rieke et al. (2008) and Casagrande et al. (2012) by using accurate determinations for the stellar temperatures and matching to the temperature of the Sun. The second issue is that the accuracy of the absolute measurements of the Sun was overestimated in the early 21st century, and there was a controversy in the solar community regarding the correct values (Gueymard 2018). This issue has been resolved, and the spectrum of Coddington et al. (2021) has the advantages of a very accurate absolute calibration and realistic spectral features. In Section 3.7 we took advantage of these advances to generate an accurate calibration using Johnson’s (1965) method.

In addition to their foundation in the flux from a real and well-behaved star, solar-based calibrations have the advantages that the stars tend to be significantly less variable than A-stars, and once they pass 1 Gyr in age, they rotate slowly. Young G-stars have debris disk excesses at  $24 \mu\text{m}$  at a rate similar to that of A-stars, but the incidence goes down dramatically for ages  $>500$  Myr (Gáspár et al. 2013), so there is a significant advantage in using them well into the middle infrared compared with A-stars (which do not stay on the main

sequence past  $\sim 1$  Gyr). The incidence of hot near-infrared excesses appears to be lower than for A-stars (3/16 for F9–G8 dwarfs; Absil et al. 2013; Núñez et al. 2017). Spectra similar in quality to those used by Kovtyukh et al. (2003) for their LDR temperature determinations can be obtained in a few hours down to about 15th–16th magnitude with modern echelle spectrographs.<sup>22</sup> For nearby stars, where extinction is low, temperatures can also be determined from the  $V - K_S$  color calibrated against the LDR temperatures as in Figure 4. Equation (6) is the fit to the data:

$$T_{\text{eff}} = -1149.5 \times (V - K_S) + 7580.7. \quad (6)$$

A more general treatment including the dependence on  $\text{Fe}/\text{H}$  and  $\log g$  can be found in Masana et al. (2006).

Temperatures are also available, of course, from accurate spectral types. With suitable temperature measurements, solar-type stars are suitable for accurate calibration for the visible through the mid-infrared. However, a liability is that their absorption features have significant dependence on temperature/spectral type and  $\log g$  (and some on metallicity), and atmospheric models generally do not predict their SEDs to sufficient accuracy to support fully the absolute calibration. In view of these issues, in Appendix D we provide an accurate SED of the Sun, which can be used as a metric to test atmospheric models and their predicted SEDs, and, of course, can support accurate calibration using stars close in characteristics to the Sun. These issues will be explored in a future paper (G. Rieke, in preparation).

#### 4.3.3. White Dwarfs

White-dwarf stars have been widely used for calibration in the ultraviolet and visible. They tend to be too faint to be generally useful in this role in the infrared, and in fact are not nearly as well studied there as A- and G-stars are. A significant advance is the models in Bohlin et al. (2020) and the demonstration in Bohlin et al. (2022) and in this paper that the three prime calibrator WDs follow the Sirius spectrum through IRAC Band 4, at  $8 \mu\text{m}$ . However, the errors for the two longer wavelength IRAC bands are large enough that this agreement is not well established beyond  $5 \mu\text{m}$ . Furthermore, it is not clear how general this result is. The incidence of mid-infrared excesses due to circumstellar dust may be significant (Farihi 2016), and the excesses are variable (Swan et al. 2020). Background limits for ground-based telescopes make use of WD calibrators in the deep thermal infrared (i.e.,  $4\text{--}30 \mu\text{m}$ ) impossible with existing telescopes.<sup>23</sup>

For the near term, it is clear that A- and G-stars have to carry the largest role in infrared calibration, specifically Am, Ap, and G-stars of age  $>1$  Gyr. However, the smooth continua, high temperatures (hence well-constrained visible-to-infrared SEDs), and lack of variability make WDs the best choice to bridge the accurate calibration now available from the visible into the near-infrared.

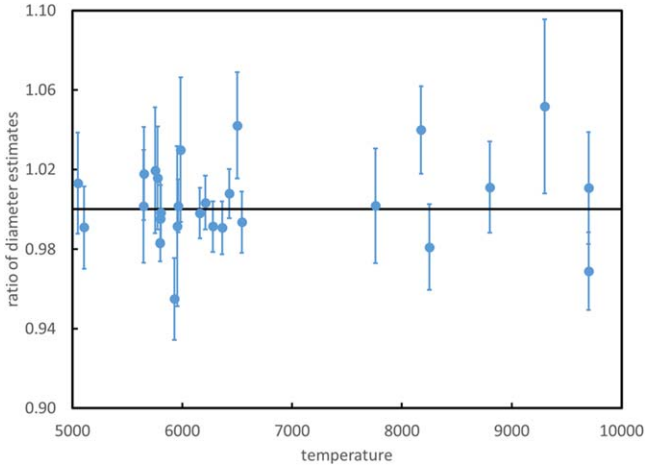
#### 4.3.4. Interstellar Reddening

For bright calibration stars within the Local Bubble (i.e.,  $<100$  pc distant), extinction and interstellar reddening are very small, in fact virtually insignificant in the infrared (Reis et al. 2012).

<sup>22</sup> For example, the UV-Visual Echelle Spectrograph on the Very Large Telescope.

<sup>23</sup> See <https://www.gemini.edu/instrumentation/niri/capability#Sensitivity>.





**Figure 5.** Ratio of stellar diameters determined by the IRFM to the limb-darkened diameters measured interferometrically, as a function of  $T_{\text{eff}}$ .

However, faint G- and A-stars must be at greater distance, and their SEDs will typically be affected to levels well outside the overall calibration accuracy. However, the effects of interstellar reddening and small differences in stellar temperature tend to be degenerate. As an example for solar-type stars, if  $T_{\text{eff}}$  from spectral classification is compared with that from Equation (6), extinction will result in a lower value from Equation (6) than is intrinsic to the star. An estimate of the amount of reddening is the level that adjusts  $V - K_S$  to bring the two estimates into agreement. If the assumed extinction is adjusted to make the  $V - K_S$  match that expected for the temperature, we show in Appendix E that the resulting SED will be sufficiently accurate to preserve the absolute calibration. Although this approach may sound artificial, it is no different from what is necessary with any other type of calibration star where extinction is an issue: one must use one band (in this case  $K_S$ ) to normalize the SED and another band (in this case  $V$ ) to constrain the amount of extinction.

#### 4.4. Accurate Stellar Diameters from the IRFM Using Just a Spectral Type and $K_S$ Magnitude

A final test for systematic errors is to see if the calibration can be used to determine accurate angular diameters of main-sequence stars, i.e., using the IRFM, Equation (2) solved for  $\theta$ . The method is a good test because it is simple and direct, requiring only spectral classification of the stellar type and a measurement in the  $K$  band. Previous work along similar lines has been reported by Casagrande et al. (2014), Boyajian et al. (2014), and Adams et al. (2018). Boyajian et al. (2014) fit measured diameters against a wide selection of photometric colors to determine purely empirical relations. They found the diameters can be derived in this way to within random errors of  $\lesssim 5\%$ . Adams et al. (2018) expanded on this approach; the smallest errors in their approach are when  $H$ - and/or  $K$ -band measurements are included, for which they quote uncertainties of 1.8%–2.9%. Casagrande et al. (2014) were concerned with the larger issues of not only confirming that the IRFM diameters agree with those found by interferometry, but also improving the scale of effective temperatures. They depended on visible as well as infrared photometry to determine the bolometric flux of their target stars. Taking their

Tycho2-SAAO determinations, they returned IRFM diameters about  $\sim 2\%$  larger than the interferometric ones with an rms scatter of 3%. Obviously, all of these methods will have some sensitivity to interstellar reddening, depending on the visible bands selected.

##### 4.4.1. Recipe

We describe here all of the steps we use to determine stellar diameters from  $K$  magnitudes and a temperature determined spectroscopically. The diameter of a star, from Equation (2), can be expressed as

$$\theta = 2 \times \sqrt{F(\lambda_0) / (\pi \times \phi(T_{\text{eff}}, g, \text{Fe}/\text{H}, \lambda_0))} \quad (7)$$

where we now can insert the final calibration into Equation (5):

$$F(\lambda_0) = 6.4325 \times 10^{-14} \times 10^{(-0.4 \times \text{mag})} W \text{cm}^{-2} \mu\text{m}^{-1}, \quad (8)$$

where mag is the magnitude in IRAC Band 1 (and  $\lambda_0$  is its wavelength). This is useful if one has IRAC Band 1 measurements; an outstanding example is the observation in this band of the entire Kepler field (Werner et al. 2021). However, for demonstration purposes, we will use  $K$ -band photometry, since in principle this can be obtained for any star of interest. Band 1 magnitudes can be determined from  $K$  ones as in Rieke et al. (2022), if one has the intrinsic  $V - K_S$  color difference, that is:

$$\begin{aligned} K_S - \text{IRAC1} = & -0.001228 \times x^4 + 0.014382 \\ & \times x^3 - 0.042158 \times x^2 + 0.057587 \times x, \end{aligned} \quad (9)$$

where  $x = V - K_S$ .

Obtaining the intrinsic value requires correction for reddening. Although that effect can largely be ignored for stars within the Local Bubble, the method is not general with that requirement. Instead, the intrinsic color difference can be estimated from the stellar temperature,  $T_{\text{eff}}$ :

$$\begin{aligned} K_S - \text{IRAC1} = & 4.884626 \times 10^{-23} T_{\text{eff}}^6 - 2.370036 \\ & \times 10^{-18} T_{\text{eff}}^5 + 4.734119 \times 10^{-14} T_{\text{eff}}^4 \\ & - 4.979066 \times 10^{-10} T_{\text{eff}}^3 + 2.905681 \\ & \times 10^{-6} T_{\text{eff}}^2 - 0.00892238 T_{\text{eff}} + 11.30719, \end{aligned} \quad (10)$$

where the association of temperature with spectral type can be taken from Pecaut & Mamajek (2013). The  $K_S$  magnitude needs also to be corrected for extinction, which can be done from the stellar colors. That is, with only a spectral type and a measurement at  $K$  and another shorter-wavelength color, it is possible to determine a stellar angular diameter which, as we demonstrate in the next section, has an accuracy rivaling the interferometric ones. This derivation is valid only for main-sequence stars of temperature between 4800 and 10,000 K, of roughly solar metallicity. Pursuing this method over a broader range of spectral types and metallicities is beyond the scope of this paper.

##### 4.4.2. Demonstration of Approach

Since the stars with accurately measured diameters are bright, usually too bright for unsaturated 2MASS photometry, we use heritage photometry transformed to  $K_S$ , as in Rieke et al. (2022). The calculations use the simplified approach



**Table 11**  
Comparison of Stellar Diameter Measurements

| HR <sup>a</sup> | Seis-<br>mology<br>mas | Error<br>mas | Reference <sup>b</sup> | IR Int.<br>$\theta_{LD}$<br>mas | Error<br>mas | Ref. <sup>b</sup> | Vis. Int.<br>$\theta_{LD}$<br>mas | Error<br>mas | Reference <sup>b</sup> | This<br>Work <sup>c</sup><br>mas |
|-----------------|------------------------|--------------|------------------------|---------------------------------|--------------|-------------------|-----------------------------------|--------------|------------------------|----------------------------------|
| 2943            | 5.375                  | 0.027        | 1                      | 5.448                           | 0.053        | 5                 | 5.446                             | 0.054        | 9                      | 5.492                            |
| "               | 5.36                   | 0.07         | 2                      | 5.39 <sup>d</sup>               | 0.03         | 2                 | 5.43                              | 0.07         | 10                     |                                  |
| "               |                        |              |                        | 5.401 <sup>d</sup>              | 0.031        | 6                 | 5.46                              | 0.08         | 10                     |                                  |
| "               |                        |              |                        |                                 |              |                   | 5.406                             | 0.08         | 11                     |                                  |
| 6060            | 0.669                  | 0.002        | 3                      | 0.780                           | 0.017        | 7                 | 0.669                             | 0.007        | 12                     | 0.674                            |
| "               | ...                    | ...          | ...                    | ...                             | ...          | ...               | 0.676                             | 0.006        | 13                     |                                  |
| 7503            | 0.547                  | 0.0035       | 4                      | 0.554                           | 0.011        | 8                 | 0.531 <sup>d</sup>                | 0.004        | 14                     | 0.553                            |
| "               |                        |              |                        |                                 |              |                   | 0.539                             | 0.007        | 12                     |                                  |
| 7504            | 0.496                  | 0.0031       | 4                      | 0.513                           | 0.012        | 8                 | 0.484 <sup>d</sup>                | 0.005        | 14                     | 0.502                            |
| "               |                        |              |                        |                                 |              |                   | 0.490                             | 0.006        | 12                     |                                  |

**Notes.**<sup>a</sup> HR 2943 = Procyon, HR 6060 = 18 Sco, HR 7503 = 16 Cyg A, HR 7504 = 16 Cyg B.<sup>b</sup> References: 1: Yıldız et al. (2019); 2: Chiavassa et al. (2012); 3: Li et al. (2012); 4: Metcalfe et al. (2012); 5: Kervella et al. (2004); 6: Aufdenberg et al. (2005); 7: Boyajian et al. (2012); 8: Boyajian et al. (2013); 9: Mozurkewich et al. (2003); 10: Nordgren et al. (2001); 11: Baines et al. (2021); 12: White et al. (2013); 13: Bazot et al. (2011); 14: Karovicova et al. (2022).<sup>c</sup> See Section 4.4.<sup>d</sup> Three-dimensional limb darkening model employed.

**Table 12**  
Diameter Estimates

| HR   | $T_{\text{eff}}$<br>K | Reference <sup>a</sup> | IRAC1<br>mag | Reference <sup>a</sup> | est. $\theta_{LD}$<br>mas | $\theta_{LD}$<br>mas | Error<br>mas | Reference <sup>a</sup> | Ratio | Error |
|------|-----------------------|------------------------|--------------|------------------------|---------------------------|----------------------|--------------|------------------------|-------|-------|
| 508  | 5662                  | 2                      | 4.565        | 7,8                    | 0.519                     | 0.531                | 0.006        | 21                     | 1.022 | 0.023 |
| 937  | 5963                  | 1                      | 2.617        | 6                      | 1.240                     | 1.246                | 0.007        | 22                     | 1.005 | 0.013 |
| 1136 | 5182                  | 3                      | 1.355        | 9,10,11                | 2.388                     | 2.394                | 0.014        | 23                     | 1.002 | 0.021 |
| 1543 | 6422                  | 2,3                    | 2.046        | 6                      | 1.552                     | 1.526                | 0.004        | 22                     | 0.983 | 0.012 |
| 2007 | 5776                  | 1                      | 4.373        | 12,13,14               | 0.562                     | 0.572                | 0.009        | 21                     | 1.019 | 0.026 |
| 2491 | 9850                  | 5                      | -1.395       | 15                     | 6.195                     | 6.04                 | 0.014        | 24                     | 0.975 | 0.020 |
| 2943 | 6562                  | 2,3                    | -0.706       | 6                      | 5.454                     | 5.448                | 0.053        | 25                     | 0.999 | 0.014 |
| 4357 | 8175                  | 4                      | 2.217        | 9,10,11                | 1.276                     | 1.329                | 0.009        | 22                     | 1.041 | 0.022 |
| 4540 | 6055                  | 1                      | 2.261        | 6                      | 1.449                     | 1.433                | 0.006        | 22                     | 0.989 | 0.013 |
| 4983 | 5954                  | 1                      | 2.813        | 7,8,11, 14             | 1.133                     | 1.127                | 0.040        | 22                     | 0.994 | 0.040 |
| 5107 | 8800                  | 4                      | 3.046        | 8,11,13,16             | 0.842                     | 0.852                | 0.009        | 22                     | 1.012 | 0.023 |
| 5404 | 6220                  | 3                      | 2.777        | 6                      | 1.127                     | 1.109                | 0.007        | 22                     | 0.984 | 0.013 |
| 5447 | 6650                  | 2,3                    | 3.459        | 11,16,17               | 0.796                     | 0.841                | 0.013        | 22                     | 1.057 | 0.027 |
| 5933 | 6302                  | 2,3                    | 2.592        | 6                      | 1.219                     | 1.217                | 0.005        | 22                     | 0.998 | 0.013 |
| 6060 | 5799                  | 1                      | 3.970        | 18                     | 0.674                     | 0.671                | 0.005        | 26                     | 0.995 | 0.021 |
| 6623 | 5500                  | 3                      | 1.702        | 8,11                   | 1.971                     | 1.934                | 0.01         | 27                     | 0.981 | 0.021 |
| 7061 | 6388                  | 2,3                    | 3.009        | 6                      | 0.999                     | 1.000                | 0.006        | 22                     | 1.001 | 0.013 |
| 7235 | 9700                  | 4                      | 2.843        | 11,19                  | 0.886                     | 0.895                | 0.017        | 22                     | 1.010 | 0.028 |
| 7503 | 5803                  | 1                      | 4.399        | 8,12,20                | 0.553                     | 0.543                | 0.011        | 21                     | 0.981 | 0.021 |
| 7504 | 5752                  | 1                      | 4.622        | 7,11,12                | 0.502                     | 0.496                | 0.012        | 21                     | 0.989 | 0.031 |
| 7924 | 5795                  | 6                      | -1.562       | 6                      | 8.621                     | 8.511                | 0.038        | 28                     | 0.987 | 0.013 |
| 7957 | 4858                  | 1                      | 1.196        | 7,11                   | 2.663                     | 2.650                | 0.04         | 29                     | 0.995 | 0.021 |
| 8314 | 5970                  | 2                      | 4.528        | 12,19                  | 0.514                     | 0.530                | 0.015        | 21                     | 1.032 | 0.036 |
| 8585 | 9050                  | 4                      | 3.719        | 7,8                    | 0.603                     | 0.610                | 0.022        | 22                     | 1.039 | 0.042 |
| 8728 | 8250                  | 4                      | 0.959        | 9,10,13,16             | 2.268                     | 2.228                | 0.020        | 28                     | 0.982 | 0.022 |
| 8969 | 6239                  | 2                      | 2.825        | 19                     | 1.100                     | 1.082                | 0.009        | 22                     | 0.983 | 0.021 |
| 8984 | 7766                  | 4                      | 3.986        | 8,12                   | 0.579                     | 0.581                | 0.012        | 21                     | 1.003 | 0.029 |

**Note.**<sup>a</sup> Notes: 1: LDR, Kovtyukh et al. (2003); 2: Malyuto & Shvelidze (2011); 3: Muñoz Bermejo et al. (2013); 4: from spectral type & Pecaut & Mamajek (2013); 5: Bohlín et al. (2017); 6: this work; 7: Alonso et al. (1998); 8: Kidger & Martín-Luis (2003); 9: Carter (1990); 10: Glass (1974); 11: Johnson et al. (1966); 12: 2MASS; 13: Bouchet et al. (1991); 14: Allen & Cragg (1983); 15: Rieke et al. (2022); 16: UKIRT standards; 17: Aumann & Probst (1991); 18: Casagrande et al. (2014); 19: K. Su & G. Rieke, unpublished; 20: Campins et al. (1985); 21: Boyajian et al. (2013); 22: Boyajian et al. (2012); 23: Thévenin et al. (2005); 24: Kervella et al. (2003); 25: Davis et al. (2011); 26: Kervella et al. (2004); 27: Mozurkewich et al. (2003); 28: Baines et al. (2014, 2018); 29: Kervella et al. (2017); 29: Di Folco et al. (2004).

**Table 13**  
Zero-points for Photometric Bands

| Band  | $\lambda_{\text{pivot}}$<br>$\mu\text{m}$ | Zero-point<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | Jy    | C<br>$\text{W cm}^{-2} \mu\text{m}^{-1}$ | $\alpha$<br>$\lambda$ in $\mu\text{m}$ | $\Delta^a$<br>mag  |
|-------|-------------------------------------------|---------------------------------------------------|-------|------------------------------------------|----------------------------------------|--------------------|
| V     | 0.5511 <sup>b</sup>                       | $3.578 \times 10^{-12\text{c}}$                   | 3622  | ...                                      | ...                                    | <sup>d</sup>       |
| J     | 1.243 <sup>c</sup>                        | $3.075 \times 10^{-13}$                           | 1621  | $6.70587 \times 10^{-13}$                | −3.58771                               | −0.014             |
| H     | 1.653 <sup>c</sup>                        | $1.1274 \times 10^{-13}$                          | 1027  | $6.0950 \times 10^{-13}$                 | −3.35764                               | 0.024 <sup>d</sup> |
| $K_S$ | 2.1603 <sup>c</sup>                       | $4.225 \times 10^{-14}$                           | 658   | $7.75216 \times 10^{-13}$                | −3.7773                                |                    |
| W1    | 3.368 <sup>f</sup>                        | $7.886 \times 10^{-15}$                           | 298   | $8.49388 \times 10^{-13}$                | −3.85349                               | 0.030 <sup>g</sup> |
| W2    | 4.618 <sup>f</sup>                        | $4.225 \times 10^{-14}$                           | 165.7 | $8.68374 \times 10^{-13}$                | −3.86997                               | −0.003             |

**Notes.**

<sup>a</sup> Adjustment to put zero-point on Sirius scale; the tabulated values should be subtracted from the natural system magnitudes.

<sup>b</sup> Calculated from response curve in Bessell & Murphy (2012).

<sup>c</sup> Assuming that the V magnitude of Vega is 0.033.

<sup>d</sup> Prime calibration band.

<sup>e</sup> Calculated from 2MASS response curves from Cohen et al. (2003), available at [https://irsa.ipac.caltech.edu/data/2MASS/docs/releases/allsky/doc/sec6\\_4a.html](https://irsa.ipac.caltech.edu/data/2MASS/docs/releases/allsky/doc/sec6_4a.html)

<sup>f</sup> Calculated using relative response function from Wright et al. (2010), available at [https://wise2.ipac.caltech.edu/docs/release/allsky/expsup/sec4\\_4h.html#WISEZMA](https://wise2.ipac.caltech.edu/docs/release/allsky/expsup/sec4_4h.html#WISEZMA).

<sup>g</sup> Value from Pecaut & Mamajek (2013) corrected to ALLWISE calibration as described at [https://wise2.ipac.caltech.edu/docs/release/allwise/expsup/sec5\\_3e.html](https://wise2.ipac.caltech.edu/docs/release/allwise/expsup/sec5_3e.html).

discussed in the previous section. Different diameter determinations sometimes differ by significantly more than suggested by the quoted errors (Karovicova et al. 2022). Therefore, as an initial demonstration, we show results in Table 11 for stars with multiple diameter measurements, which suggests that this method is capable of returning diameters of similar accuracy to the other approaches. Not only is the performance achieved with a simple formulation that can be carried out with only an accurate spectral type and  $K$  magnitude, but there is no fitting: the correct diameters are returned on physical principles.

Results for a larger sample of stars are provided in Table 12 and plotted in Figure 5. Our test sample consists of the stars that *both* have accurate diameters measured through interferometry and *also* have heritage photometry of sufficient quality (and number of measurements) to allow determining an accurate  $K_S$  magnitude (expected error <1.5%). No other selections were applied. Diameters measured interferometrically require knowledge of the limb darkening, which can be a source of uncertainty (Karovicova et al. 2022). This effect is significantly smaller in the infrared than in the visible, so we give priority to stars with diameters measured in the  $H$  or  $K$  bands.

The sources for the stellar temperatures are indicated in the notes, as are the sources for the  $K$ -band photometry and the stellar diameters. We convert the  $K_S$  photometry to IRAC Band 1 (see Rieke et al. 2022), and we determine the emissivity in that band as described in Appendix C. The errors are the RSS combination of the quoted errors in the limb-darkened diameters and 2% for the diameters from the IRFM, the latter based on estimates of the contributions of the photometric errors and the errors in the temperatures.<sup>24</sup> The ratios are indistinguishable from unity within the errors. In fact, the average of the ratios of the interferometrically measured to IRFM estimated diameters is  $1.002 \pm 0.004$ , and the median is 0.998. The reduced  $\chi^2$  relative to unity is 0.83.

Clearly, the demonstration is a success; there is no indication of a systematic error in the calibration. In addition, this method should allow accurate diameters to be measured through the IRFM for any main-sequence stars, regardless of distance, with

accurate spectral types and (extinction corrected)  $K_S$ -band magnitudes.

## 5. Conclusion

We provide an absolute calibration of the A0-type star Sirius to an accuracy of 0.6% in both the infrared and visible. The result is in close agreement in both bands, validating this spectrum as a universal template for absolutely calibrated measurements from the visible throughout the mid-infrared. We therefore propose Sirius to define the stellar photometric system zero-point, replacing Vega, which over the past 30 yr has been revealed to have multiple issues in this role. We test and find no evidence for systematic errors in the calibration.<sup>25</sup> We show that the proposed infrared calibration can be used to estimate stellar diameters from first principles to a high level of accuracy, using as inputs only a spectral type and  $V$ - and  $K_S$ -band magnitudes.

This calibration can be transferred accurately to faint stars as described by Bohlin & Gilliland (2004) for the visible and Rieke et al. (2022) and Su et al. (2022) for the infrared. From there, it must be conveyed by secondary standards around the sky. We evaluate the three stellar types commonly proposed as secondary standards. We show that normal A-stars are not suitable for this use to the necessary accuracy, although Am and Ap stars may be significantly better in this application. Solar-type stars older than  $\sim 1$  Gyr are in principle satisfactory, although only if the combination of the effects of stellar temperature and interstellar reddening can be determined accurately. White dwarfs are the best suited to provide a uniform calibration from the visible to the near-infrared but are too faint and insufficiently explored for use at longer infrared wavelengths.

To assist in applying this calibration, we have set the CALSPEC Sirius spectrum to the average level determined by our visible and infrared calibrations, and we include it in Appendix D in electronic format. Appendix D also includes an accurate SED for the Sun in electronic format to assist in applying G-star-based calibrations. These SEDs, along with

<sup>24</sup> The stars discussed in Section 3.6 are included in the figure, with the errors from the IRFM set to 1.2% in recognition of the more direct and accurate IRAC Band 1 measurements of them.

<sup>25</sup> This result would very usefully be checked against new absolute measurement campaigns, for which our result provides a solid foundation.

those of white-dwarf stars available in the CALSPEC database, are the basic raw material for extending the calibration to spectra of other objects.

Although our photometric calibration has been centered on two bands,  $V$  and  $K_S$ , it can be expanded to other photometric bands readily. The most accurate results will be obtained by convolving the SED of the standard stars and the target object with the system relative response function and comparing the modeled signals. However, it is also useful to have a set of fiducial wavelengths and zero-points, as illustrated in Table 13. Extending the calibration in the visible region should be done by reference to the CALSPEC database. The issues with the A0V star zero-point discussed in Section 4.3.1 are significantly diminished in the infrared, enabling a reasonably accurate general approach. We start from the zero-point calibrated for the 2MASS  $K_S$  band. Since there are small offsets from zero color for A0V stars in 2MASS, we use the standard colors from Pecaute & Mamajek (2013) to estimate corrections to the 2MASS magnitudes to put them on a consistent basis for this zero-point. The table also includes the parameters for power-law fits to the continua within each filter, of the form  $F_\lambda = C\lambda^\alpha$  to allow for adjusting the zero-point to other choices of fiducial wavelength, e.g., effective, nominal, or isophotal (see Rieke et al. 2008 for definitions and discussion).

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## Appendix A Spectral Templates for $\alpha$ Boo and $\beta$ Gem

The CWW spectrum of the K1.5III star  $\alpha$  Boo (Cohen et al. 1996) appears to be the noisiest in the 12–20  $\mu\text{m}$  range where MSX bands C and D lie. However, the noise appears at a level where it should average out reasonably well over the broad photometric bands used in the MSX measurements. To test this conclusion, we have compared the spectra integrated over each of the A, C, and D bands with results using the Engelke et al. (2006) spectrum, finding the first two to agree well but the last to be 1.72% higher. Given these issues, we constructed a new spectral component from 10–20  $\mu\text{m}$  based on the Spitzer IRS spectrum of HD 214873. Although this star is nominally of type K3III, its SiO feature matches that of  $\alpha$  Boo closely, and it has a high-quality spectrum from Sloan et al. (2015). The shifts in bands C and D with this alternative spectrum were insignificant ( $\leq 0.4\%$ ), showing that the original Cohen et al. (1996)/CWW spectrum is an accurate comparison for the MSX observations. However, it should be used with caution in other applications, e.g., at higher spectral resolution than the MSX photometric bands.

The  $\beta$  Gem template used in Price et al. (2004a) from Cohen et al. (1995) is of low quality; the replacement template proposed by Engelke et al. (2006) is improved but still has some issues. The Cohen et al. (1995) spectrum is based beyond  $\sim 12 \mu\text{m}$  on an analytic approximation, the “Engelke Function,” and this approach was also used to help stitch together spectral segments at shorter wavelengths. For comparison (but obviously not used for the previous MSX analysis), the improved spectrum by Engelke et al. (2006) replaced this function with a measured spectrum. However, this spectrum is not of  $\beta$  Gem itself, which was not observed spectroscopically

**Table 14**  
Properties of Stand-in Stars for  $\beta$  Gem<sup>a</sup>

| Star         | $T_e$ | $\log g$ | Fe/H  | $\log(L/L_\odot)$ |
|--------------|-------|----------|-------|-------------------|
| $\beta$ Gem  | 4998  | 3.13     | 0.17  | 1.58              |
| HR 6348      | 4989  | 3.04     | 0.17  | 1.63              |
| $\delta$ Dra | 4952  | 2.90     | −0.08 | 1.77              |

**Note.**

<sup>a</sup> All parameters from Luck & Heiter (2007) based on high-resolution ( $\lambda/\Delta\lambda \sim 60,000$ ), high signal-to-noise ( $\sim 150$ ) spectroscopy from 475–685 nm.

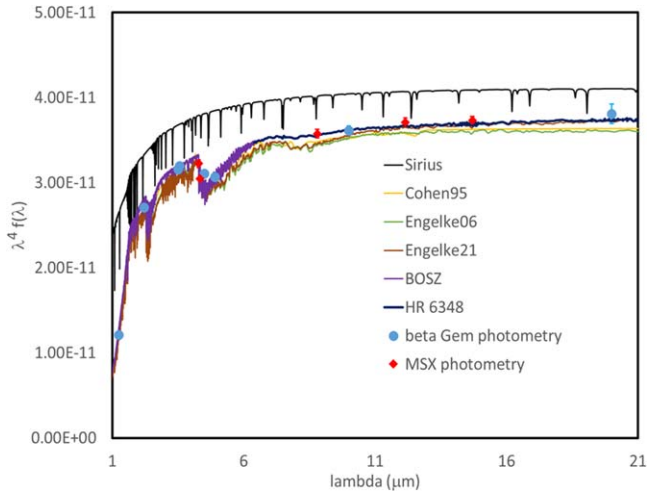
with ISO-SWS, but instead of other stars. Both of these,  $\theta$  Cen and  $\delta$  Dra, have full ISO SWS spectra but they have significant noise artifacts between 12 and 20  $\mu\text{m}$ . ISO Cam CVF data were used also, but calibrating them to the necessary level of accuracy has proven challenging (Engelke et al. 2004). The signal to noise in the resulting spectrum is modest and beyond 12.9  $\mu\text{m}$ , covering a small piece of the MSX C band and all of the D band, the net spectrum shows departures from Rayleigh–Jeans behavior compared with the other template stars. For example, despite the far weaker SiO feature in the  $\beta$  Gem spectrum, it falls 1%–1.5% more slowly between 10.5 and 13  $\mu\text{m}$  than do the other templates.

The issues mentioned in Section 3.2.1 have led to our putting together a new template spectrum for  $\beta$  Gem. To do so, we will use the Spitzer IRS spectrum of HR 6348 as a stand-in for the 5–20  $\mu\text{m}$  range (and to 35  $\mu\text{m}$ ), since there is no spectrum available for  $\beta$  Gem itself. This star is very closely matched to  $\beta$  Gem, as shown in Table 14, in fact significantly more closely than is  $\delta$  Dra, which is the primary stand-in star for the Engelke et al. (2006) template.<sup>26</sup> HR 6348 has a very high-quality IRS spectrum, and was the type K0III “truth spectrum” for the calibration of the Spitzer IRS (Sloan et al. 2015). We also use a theoretical spectrum from the library of models described by Bohlin et al. (2017), adopting  $T_e = 4850$  K, Fe/H = 0.0, and  $\log g = 2.85$  (and interpolating among the models).

Our approach is to assemble all of the high-quality photometry of  $\beta$  Gem between 2 and 25  $\mu\text{m}$ , *except* for the MSX values. We use measurements that are directly relative to Sirius within the same program so far as possible, since doing so can reduce errors caused by passing through photometric systems and networks of intermediate standard stars. The one exception is that we take the measurement of Sirius at the N band from van der Blik et al. (1996), since the star was not included by Hammersley et al. (1998). We use the photometry to assemble the template spectrum relative to the reference spectrum for Sirius. This spectrum is then compared with the MSX absolute photometry to determine any offsets relative to the measurements of Sirius, thus generating offset terms similar to those in Table 4.

This comparison requires accurate spectral templates for both stars; for Sirius, we use the current (2020) model in CALSPEC (Bohlin et al. 2020). It and our template for  $\beta$  Gem are shown in Figure 6. We discuss the derivation of the latter template in more detail. The BOSZ model spectrum for the star agrees well with that from Engelke et al. (2006) at wavelengths shorter than  $\sim 6 \mu\text{m}$ , except for a narrow peak in the Engelke model near 4.2  $\mu\text{m}$ . Comparing with other stellar spectra, the

<sup>26</sup> Some of the parameters in the comparisons differ from more common values, e.g.,  $T_e$ , but we prefer the reference cited because it uses identical methods on the stars.



**Figure 6.** Template spectrum for  $\beta$  Gem and measurements relative to Sirius. The spectrum of Sirius is from CALSPEC (Bohlin et al. 2020). Four templates are shown: (1) from Cohen et al. (1995); (2) from Engelke et al. (2006); (3) an updated version by Engelke; and (4) as derived in this paper combining the BOSZ model ( $\lambda < 6.5 \mu\text{m}$ ) and the spectrum of HR 6348 ( $\lambda > 6.5 \mu\text{m}$ ). The blue round points are photometry relative to Sirius as described in the text. The red diamond points are measurements obtained with MSX (Price 2004), also shown relative to Sirius.

model result is to be preferred. The BOSZ model and the spectrum of HR 6348 agree very well from  $5.2 \mu\text{m}$  (the shortest wavelength in the spectrum) to  $20 \mu\text{m}$  and beyond, except that the spectrum shows a shallow SiO absorption that is missing from the model. This absorption is also apparent, although at modest signal to noise, in the Engelke model, and is expected in red giants in general. We therefore constructed the template by merging the BOSZ model spectrum and that of HR 6348 at  $6.5 \mu\text{m}$ ; since they agree over a broad spectral range around this value, the choice of merger wavelength is not critical. Finally, we normalized the complete model using the high-quality photometry relative to Sirius shown in Table 15.

## Appendix B

### Photometry of $\alpha$ Centauri A

Table 16 provides a summary of the discussion of the  $K_S$  magnitude of  $\alpha$  Cen A. There are two  $K$ -band measurements of  $\alpha$  Cen through ground-based photometry. The first is from Engels et al. (1981), using the photometer and photometric system defined by Wamsteker (1981). To derive a more accurate value at  $K$ , we used the solar-type colors in Wamsteker (1981) to correct the  $J$  and  $H$  magnitudes to equivalent  $K$  ones and averaged the three estimates in that band to find  $K = -1.477$  mag. We use the transformation in Rieke et al. (2022) to convert this value to the equivalent 2MASS one,  $K_S = -1.520$ . The error in this value is difficult to estimate because of the transformations required and also because it is a single measurement (but from the discussion in Rieke et al. 2022, should be  $\sim 2.5\%$ ).

We can derive another estimate of the  $K$  magnitude making use of the photometry reported by Thomas et al. (1973) and the DIRBE photometry from Smith et al. (2004). Thomas et al. (1973) report measurements of a number of stars including  $\alpha$  Cen relative to  $\gamma$  Cru. Most of their comparison stars, including  $\gamma$  Cru, are M giants and are likely to have significant spectral features in the near-infrared, making comparisons between different photometric systems difficult. We use the

**Table 15**  
Photometry of  $\beta$  Gem Relative to Sirius

| Wavelength ( $\mu\text{m}$ ) | Difference (mag) | Error (mag) | Reference                  |
|------------------------------|------------------|-------------|----------------------------|
| 1.25                         | 0.858            | .015        | Price et al. (2010)        |
| 1.25                         | 0.84             | .02         | Alonso et al. (1994)       |
| 1.25                         | 0.85             | 0.015       | Johnson et al. (1966)      |
| average                      | 0.851            | 0.01        |                            |
| 2.2                          | 0.258            | 0.02        | Johnson (1965)             |
| 2.2                          | 0.279            | 0.015       | Smith et al. (2004)        |
| 2.2                          | 0.271            | 0.012       | average                    |
| 3.5                          | 0.211            | 0.015       | Smith et al. (2004)        |
| 3.56                         | 0.202            | 0.01        | Su et al. (2022)           |
| 4.5                          | 0.261            | 0.01        | Su et al. (2022)           |
| 4.9                          | 0.293            | 0.015       | Smith et al. (2004)        |
| 9                            | 0.18             | 0.04        | Akari                      |
| 10.1                         | 0.18             | 0.02        | Tokunaga (1984)            |
| N (10.6 $\mu\text{m}$ )      | 0.16             | 0.02        | Hammersley et al. (1998)   |
| 10                           | 0.171            | 0.014       | average                    |
| 18                           | 0.11             | 0.04        | Akari                      |
| 20                           | 0.15             | 0.03        | Tokunaga (1984)            |
| 24                           | 0.13             | 0.12        | Engelbracht et al. (2007)  |
| 25                           | 0.19             | 0.07        | IRAS, Point Source Catalog |
| 20                           | 0.141            | 0.023       | average                    |

measurement of  $\alpha$  Car in Thomas et al. (1973); that of  $\lambda$  Vel is contaminated by a nearby star, and the rest are of M giants. We use the DIRBE photometry to measure  $\alpha$  Car and then the Thomas et al. (1973) measurement to go from that star to  $\alpha$  Cen. The result is a  $K$  magnitude of  $-1.524$  for  $\alpha$  Cen A, on the UKIRT system. In this case, the magnitude is essentially the same as that at  $K_S$ . We estimate an error of about 4%, based on the combination of errors reported by Thomas et al. (1973) and those arising from the somewhat roundabout calibration we have used.

We have derived another value from the DIRBE measurements. Because the DIRBE beam is large ( $0^\circ 7$ ), we follow the procedures described in Rieke et al. (2022) to verify that the photometry is valid. That is, we use the 2MASS data to search around each star for additional stars that could contaminate the measurement. In the case of  $\alpha$  Cen, the 2MASS data indicate the presence of a star just barely below a 1% threshold about  $17''$  from  $\alpha$  Cen. However, because it is saturated, the errors are large; the star is detected at higher weight in DENIS (DENIS Consortium 2005) 0.7 mag below the apparent 2MASS value, so the  $\alpha$  Cen photometry easily meets our conditions for use of DIRBE photometry in the far field. In the near field, we have to contend with  $\alpha$  Cen B,  $12''$  away. This star is sufficiently close that the DIRBE measurements will simply merge the signals from the two. Rieke et al. (2022) showed that the DIRBE photometry at  $2.2 \mu\text{m}$  will provide a measurement of the combined signal accurate to  $\sim 1\%$ .

We have used two approaches to estimate the contribution of the fainter star to the combined magnitude. First, the two stars have been measured individually by Engels et al. (1981), and since they are close on the sky, the relative calibrations should



**Table 16**  
Photometry of  $\alpha$  Cen A

| Band              | Magnitude | Error | Reference             | Color Rel K | Error | Reference              | $K$ mag | Error |
|-------------------|-----------|-------|-----------------------|-------------|-------|------------------------|---------|-------|
| $K_S$             | -1.524    | 0.04  | Thomas et al. (1973)  | 0.00        | ...   | ...                    | -1.524  | 0.04  |
| $K_S$             | -1.520    | 0.025 | Engels et al. (1981)  | 0.00        | ...   | ...                    | -1.520  | 0.025 |
| $K_S$             | -1.540    | 0.013 | Price et al. (2010)   | 0.00        | ...   | ...                    | -1.540  | 0.013 |
| average ( $K_S$ ) | -1.535    | 0.011 | this work             | 0.00        | ...   | ...                    | -1.535  | 0.011 |
| [3.5]             | -1.606    | 0.015 | Smith et al. (2004)   | 0.065       | 0.015 | this work              | -1.541  | 0.021 |
| [3.5]             | -1.611    | 0.015 | Price et al. (2010)   | 0.065       | 0.015 | this work              | -1.546  | 0.021 |
| L                 | -1.593    | 0.039 | Engels et al. (1981)  | 0.05        | 0.01  | Bessell & Brett (1988) | -1.543  | 0.040 |
| L                 | -1.544    | 0.05  | Thomas et al. (1973)  | 0.05        | 0.01  | Bessell & Brett (1988) | -1.494  | 0.051 |
| [24]              | -1.584    | 0.025 | Wiegert et al. (2014) | 0.045       | 0.011 | Rieke et al. (2008)    | -1.539  | 0.027 |
| average (not K)   |           |       |                       |             |       |                        | -1.540  | 0.012 |
| average (all)     | ...       | ...   | ...                   | ...         | ...   | ...                    | -1.537  | 0.008 |

be particularly good and the relative brightnesses well determined. They indicate that a fraction of 0.692 of the total at the  $K$  band comes from  $\alpha$  Cen A. The second approach is based on the diameters of the two stars in Kervella et al. (2017) and emissivities determined from the BOSZ models (Bohlin et al. 2017). The fraction of the total contributed by  $\alpha$  Cen A is 0.694 in this approach. We have used the average, 0.693. We then find  $K_S = -1.540$ . Given the additional uncertainty associated with removing the contribution of the companion star, we assign an error of 1.3% rather than the 1% typical of DIRBE measurements in general.

The weighted average of these three determinations of the  $K_S$  magnitude is  $-1.537 \pm 0.008$ .

To provide an independent check, we proceed to the longer wavelength bands. We will include all of the measurements of reasonably high signal to noise (e.g., error  $< 5\%$ ) and where we can derive a consistent calibration; Table 16 provides a summary. We start by evaluating the DIRBE  $3.5 \mu\text{m}$  measurements. To do so with an adequate number of comparison stars, we take stars brighter than 0 magnitude at  $L$  from the UKIRT bright standards plus Carter (1990) and van der Bliik et al. (1996). We use the transformations in Carter (1990) to put the ESO standards on the SAAO system and then confirm that this system is very close to the UKIRT one. Where the same star has been measured by more than one source, we average the results. We find a small color term in comparing the results:

$$\frac{\text{DIRBE}_{3.5 \mu\text{m}}}{\text{ground}_L} = -0.047(J - K) + 1.0311, \quad (\text{B1})$$

where we take the ratio of flux densities, assuming a zero-point of 284 Jy for the ground-based measurements. We then find an rms scatter of 1.8% for these ground-based measurements relative to the DIRBE results. For  $\alpha$  Cen A, we correct for the neighboring star as for  $K$ , assuming a standard  $K - L$  color difference between  $\alpha$  Cen A itself and the cooler neighboring star. The result is in Table 6, along with the  $K - L$  color for a G2V star, i.e.,  $\alpha$  Cen A, from Equation (3) and the resulting estimate of its  $K_S$  magnitude. The errors are estimated from the analysis in Rieke et al. (2022), an assumed 1% uncertainty in  $K - L$  from Bessell & Brett (1988), and another contribution from the additional color term to the DIRBE data in Equation (3).

The corresponding values for Thomas et al. (1973) and Engels et al. (1981) reflect their measured  $K - L$ , applied to their  $K$  magnitudes (transformed to  $K_S$ ). The errors are the quadratic combination of the errors in  $K_S$  and in the  $K - L$  colors. We use the reduction by Alcione Mora reported in Wiegert et al. (2014), which included a unique Monte Carlo approach to improve the accuracy and estimate the errors.<sup>27</sup> The result is a flux density of  $30.84 \pm 0.76$  Jy, equivalent to a magnitude of  $-1.584$  (using the zero-point of Rieke et al. 2008). The error in this value is 2.5%. The expected  $K_S - [24]$  color is  $0.045 \pm 0.011$  (Rieke et al. 2008).

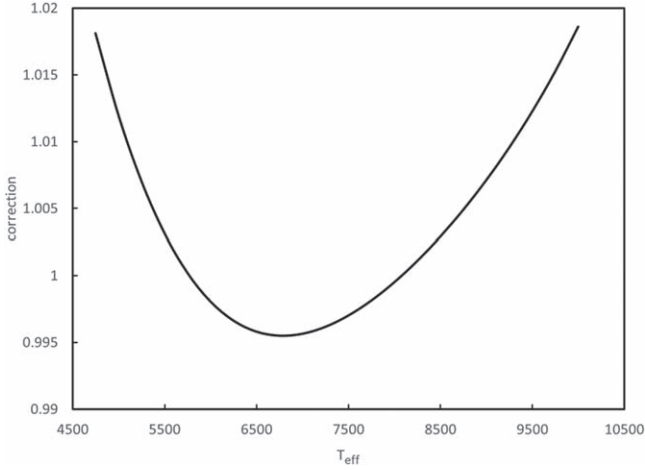
## Appendix C

### Stellar Surface Flux in IRAC Band 1

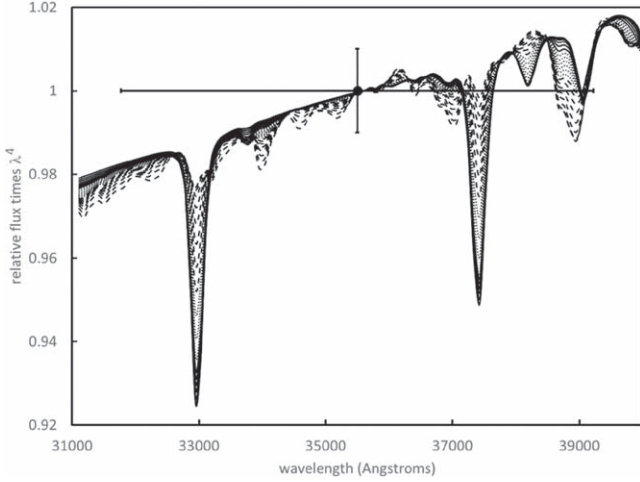
The foundation for the infrared flux method is that the behavior of the stellar surface flux at wavelength  $\lambda_0$ ,  $\phi(T_{\text{eff}}, g, \text{Fe}/\text{H}, \lambda_0)$ , should be a relatively simple function of the effective temperature,  $T_{\text{eff}}$ , surface gravity,  $g$ , and  $\text{Fe}/\text{H}$ , dominated by simple Rayleigh–Jeans behavior. Su et al. (2022) showed that the behavior of the F-stars we use for calibration relative to Sirius in IRAC Band 1 closely follows Rayleigh–Jeans behavior times the angular area of the star obtained from interferometry, an example of this behavior. We explore this phenomenon in more detail here.

First, we use the BOSZ grid of models (Bohlin et al. 2017) to explore the behavior of the surface flux as a function of stellar type. Our primary interest is in the F-type stars measured in Su et al. (2022); we use models at solar metallicity and  $\log g = 4$  to match these stars (they all have metallicity close to solar and are on the main sequence). We focus on the behavior in IRAC Band 1; from the BOSZ models, we first take out a term proportional to  $T_{\text{eff}}$  in the run of surface flux versus temperature in this band. The residuals are small, with a maximum flux predicted at a temperature of about 6750 K. That is, the correction relative to the pure Rayleigh–Jeans behavior is to slightly reduce the stellar surface flux around 6750 K and to increase it slightly at temperatures significantly below and above this value. The correction factor is shown in Figure 7, normalized to 1 for the

<sup>27</sup> The reduction reported by Liseau et al. (2013) is affected by saturation (private communication from G. Bryden, who reduced the data).



**Figure 7.** Correction to pure Rayleigh-Jeans behavior of IRAC Band 1 to obtain accurate stellar surface flux estimates. Over this temperature range, the total amplitude is only  $\sim 2\%$ .



**Figure 8.** BOSZ spectra (normalized by  $1/T_{\text{eff}}$  plus correction term illustrated in Figure 7). The dashed lines are for  $T_{\text{eff}}$  every 250 K from 4750–6750 K, the dotted lines for 7000–8250 K, and the solid lines for 8500–10,000 K. The point with error bars shows the FWHM of IRAC Band 1, and a  $\pm 1\%$  vertical range.

Sun; it can be expressed as

$$1.113 \times 10^{-16} T_{\text{eff}}^4 - 3.761 \times 10^{-12} T_{\text{eff}}^3 + 4.929 \times 10^{-8} T_{\text{eff}}^2 - 2.885 \times 10^{-4} T_{\text{eff}} + 1.6230.$$

Figure 8 shows the BOSZ spectra of dwarf stars with  $T_{\text{eff}}$  from 4750–10,000 K, normalized by  $1/T_{\text{eff}}$  and the correction factor shown in Figure 7. Over the IRAC Band 1 bandpass, the resulting spectra are remarkably similar, showing that this band is an excellent choice for applications of the IRFM, or as in our case for inverting it to achieve a calibration.

To remain as independent of theoretical stellar models as possible, we need to anchor the IRFM empirically. Equation (1) shows that the stellar surface flux can be determined from the flux density at the wavelength in question, the bolometric flux, and  $T_{\text{eff}}$  (assuming in our case that  $g$  and Fe/H are similar to those of the Sun). All of these quantities are known accurately for the Sun; we take the flux density at IRAC Band1 ( $3.55 \mu\text{m}$ ) from Gueymard (2018), and the IAU recommends values for the other two parameters (Prsa et al. 2016). The result is  $198 \text{ W cm}^{-2} \mu\text{m}^{-1}$ . We use this value in

Equation (2) to predict the flux density from a star in IRAC Band 1. This value can be compared with the measured value for a given absolute calibration. We adjust the calibration value until the agreement is optimized over the stars measured in our program described in Su et al. (2022). Further details are provided in the main body of this paper.

## Appendix D Calibrated Spectra

Table 17 provides the CALSPEC spectrum of Sirius, renormalized to the average between our calibrations at  $V$  and  $K_S$ .

This section also provides an accurate spectrum of the Sun in Table 18. For wavelengths shorter than  $2.7 \mu\text{m}$ , we have used the results from Coddington et al. (2021). At longer wavelengths, the approach is a bit more complicated, as illustrated in Figure 9. From  $2.7$ – $3.6 \mu\text{m}$ , we have used the spectrum from Gueymard (2018), carefully normalized to join onto the shorter-wavelength one accurately (this is expedited by the good agreement of this spectrum with the Coddington et al. 2021 one across the  $2.2 \mu\text{m}$  region). At  $3.6 \mu\text{m}$ , we switch to the spectrum from Rieke et al. (2008). As explained there, the basic spectrum is from Holweger & Müller (1974), but it has been modified to match more accurately the depth of the CO absorption at  $\sim 4.5$ – $6 \mu\text{m}$ . Although the spectrum has been assembled with no reliance on photometric points, it agrees closely (within 1%) with the five high-accuracy ones available: (1)  $V$  (not shown in the figure); (2)  $K_S$ ; (3)  $H$ , derived from  $K_S$  with the small color corrections discussed in the footnote in Section 3.3; (4) IRAC Band 1, determined from  $K_S$  and the transformation equation in Rieke et al. (2022); and (5) a point at  $11 \mu\text{m}$ , to be discussed below.

Vernazza et al. (1976) reviewed absolute photometric measurements of the Sun and found that those of Siedy & Goody (1959, 1960) should be reliable in the  $10 \mu\text{m}$  region.

**Table 17**  
Calibrated Spectrum of Sirius

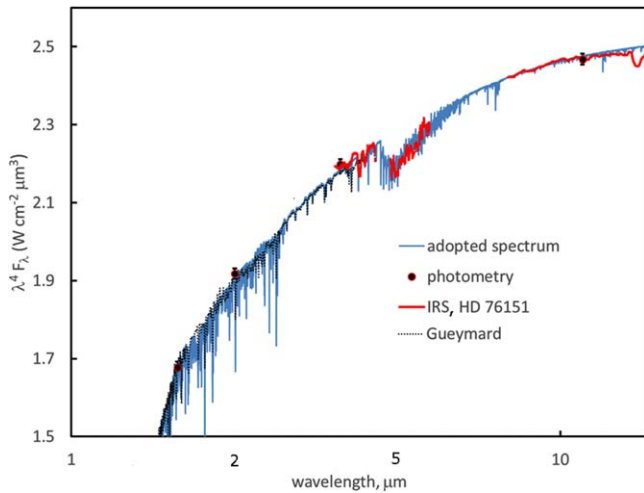
| Wavelength ( $\mu\text{m}$ ) | Flux Density ( $\text{W cm}^{-2} \mu\text{m}^{-1}$ ) |
|------------------------------|------------------------------------------------------|
| 0.11502                      | 1.438E-13                                            |
| 0.11514                      | 1.260E-13                                            |
| 0.11526                      | 1.355E-13                                            |
| 0.11538                      | 1.052E-13                                            |
| 0.1155                       | 1.327E-13                                            |
| 0.1156                       | 1.657E-13                                            |
| 0.11572                      | 1.305E-13                                            |
| 0.11584                      | 9.748E-14                                            |

(This table is available in its entirety in machine-readable form.)

**Table 18**  
Calibrated Spectrum of the Sun

| Wavelength ( $\mu\text{m}$ ) | Flux Density ( $\text{W cm}^{-2} \mu\text{m}^{-1}$ ) | Error     |
|------------------------------|------------------------------------------------------|-----------|
| 0.202                        | 0.000783                                             | 0.0000102 |
| 0.2021                       | 0.000782                                             | 0.0000102 |
| 0.2022                       | 0.000779                                             | 0.0000101 |
| 0.2023                       | 0.000777                                             | 0.0000101 |
| 0.2024                       | 0.000776                                             | 0.0000101 |
| 0.2025                       | 0.000776                                             | 0.0000101 |
| 0.2026                       | 0.00078                                              | 0.0000101 |

(This table is available in its entirety in machine-readable form.)



**Figure 9.** Adopted infrared spectrum of the Sun. As discussed in the text, the spectrum was constructed from a series of solar spectral observations and models. It agrees perfectly (within 1% with the photometry shown in the figure, extending from the  $H$  band ( $1.6 \mu\text{m}$ ) through  $15 \mu\text{m}$  (it is also in agreement with photometry at  $V$  to within 1%).

They measured the central intensity at  $8.63$ ,  $11.1$ , and  $12.02 \mu\text{m}$ , converted these values to equivalent whole-disk temperatures, and in Saiedy & Goody (1960), provided conversion factors to correct to mean fluxes (adjusting for limb darkening) for the visible disk from the values they provided. We have used the temperatures to convert each measurement to an equivalent at  $11.1 \mu\text{m}$ , corrected for limb darkening, as in Saiedy & Goody (1960), and taken the weighted average, as shown in Figure 9.

We also show two spectra. At wavelengths longer than  $7 \mu\text{m}$ , the spectrum is a composite of those of a number of solar-type stars measured with the Spitzer infrared spectrometer (IRS) and reduced as described in Rieke et al. (2008); it has been normalized to the  $11.1 \mu\text{m}$  photometry. Between  $3.5$  and  $5.7 \mu\text{m}$ , we show the spectrum of HD 76151 from Rayner et al. (2009), normalized to the photometric point at  $3.55 \mu\text{m}$ . This star is at  $T_{\text{eff}} = 5776 \text{ K}$  as measured by the line depth ratio (LDR) technique (Kovtyukh et al. 2003). It has  $\log g = 4.48$  and  $\text{Fe}/\text{H} = 0.12$  (Sousa et al. 2008). That is, its characteristics are a very close match to those of the Sun. It supports the shape of our solar spectrum around the CO fundamental band, although the use of G-stars as calibrators in this wavelength area will always be of limited accuracy.

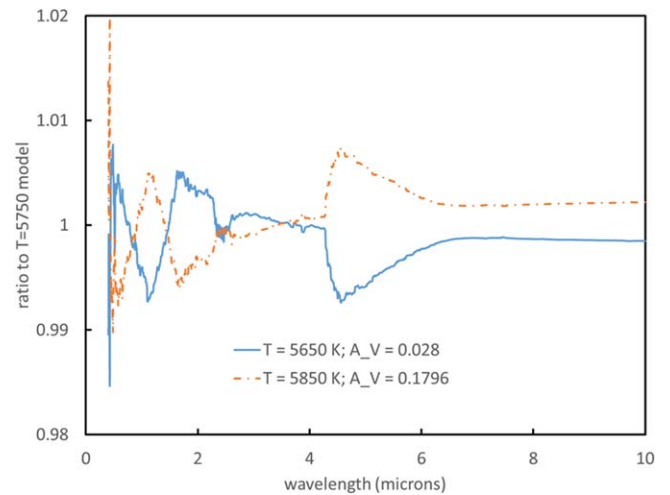
## Appendix E

### Suggested Procedure to Compensate for Reddening and for Temperature Dependence of Stellar Spectra

Determining stellar temperatures with sufficient accuracy to provide accurate templates for the visible through the mid-IR is challenging, although it may be possible using the LDR technique. However, there may be another procedure that also deals with the second source of uncertainty on the SEDs, interstellar extinction and reddening. This approach depends on the significant level of degeneracy between stellar temperature and reddening. Taking G-stars as an example, one needs a type determined spectroscopically, and accurate photometry at  $V$  and  $K_S$ . The type can be converted to  $T_{\text{eff}}$ ; we have done so using the compilation of Pecaut & Mamajek (2013). We have

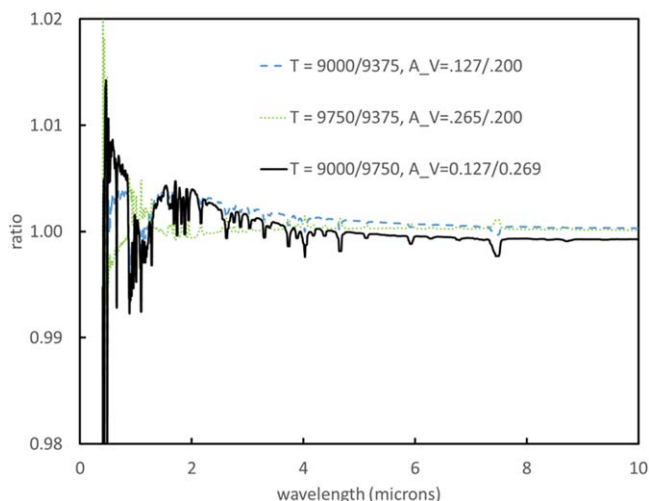
estimated the uncertainties in the temperatures by taking a suite of well-measured G-stars, obtaining all measurements of the type of each star from the compilation by Skiff (only for measurements since 1950), determining the corresponding temperatures (Pecaut & Mamajek 2013), and calculating the rms scatter of temperatures for each star. We find that the likely uncertainty in the temperature is  $\sim 50 \text{ K}$ . An alternative temperature can be estimated with similar accuracy from the relation with  $V - K_S$  in Masana et al. (2006). If the star is reddened, this latter temperature will be lower than the spectroscopic one, and the change in  $V - K_S$  required to make it agree is then a measure of the reddening.

To illustrate, we have taken interstellar extinction laws from Fitzpatrick et al. (2019) for the visible into the near-infrared and from Gordon et al. (2021) for the near-infrared and mid-infrared; we have smoothed the join between these laws with a polynomial fit. For any assumed stellar temperature, we adjust  $A_V$  to bring the shape of the resulting dereddened SED (ignoring its overall offset) into correspondence with a  $5750 \text{ K}$  SED so far as possible over the full  $0.4\text{--}30 \mu\text{m}$  range. (The SEDs are from the BOSZ compilation, Bohlin et al. 2017, for  $\log g = 4.5$  and metallicity, microturbulence, and C abundance set to the solar values). Figure 10 shows the result for temperature shifts of  $100 \text{ K}$  in either direction from  $5750 \text{ K}$ . The largest difference is at the CO fundamental band,  $\sim 4.5\text{--}5.5 \mu\text{m}$ , but generally the effects are  $< 0.5\%$ . The range in  $A_V$  is  $0.15 \text{ mag}$ . This result demonstrates that stars close in spectral type to that of the Sun can be dereddened to match the colors appropriate to their spectral types, and the dereddened SED will be a satisfactory approximation to the true one. That is, the  $\sim 50 \text{ K}$  uncertainty in  $T_{\text{eff}}$  from standard stellar type determination and the uncertainties due to interstellar reddening can be compensated. The SED can then be used to correct other photometric bands and related observations to the correct absolute calibration. Figure 11 shows a similar comparison for ideal A-stars.



**Figure 10.** Spectral shapes compared to that of a  $5750 \text{ K}$  star similar to the Sun, for two alternative stellar temperatures with the reddening adjusted to create as close a match as possible over the entire  $0.4\text{--}30 \mu\text{m}$  range (spectra from the BOSZ database, Bohlin et al. 2017, with  $\log g = 4.5$  and the other parameters set to the solar values). The ratios have been adjusted to an average of 1 for easy comparison. The ratios have been running median-averaged to smooth over absorption lines with the expectation that any use of a star for absolute calibration would interpolate over them.





**Figure 11.** Spectral shape comparisons for A-stars of various temperatures, based on models from the BOSZ database (Bohlin et al. 2017),  $\log g = 4.0$ , the other parameters set at the solar values. As in Figure 10, the reddening has been adjusted to create as close a match as possible over the entire 0.4–30  $\mu\text{m}$  range. The ratios have been adjusted to an average of 1 for easy comparison, and the ratios have been running median-averaged to smooth over absorption lines. The results apply to ideal A-stars with no temperature gradients (i.e., slow rotators) and no hot dust excesses.

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